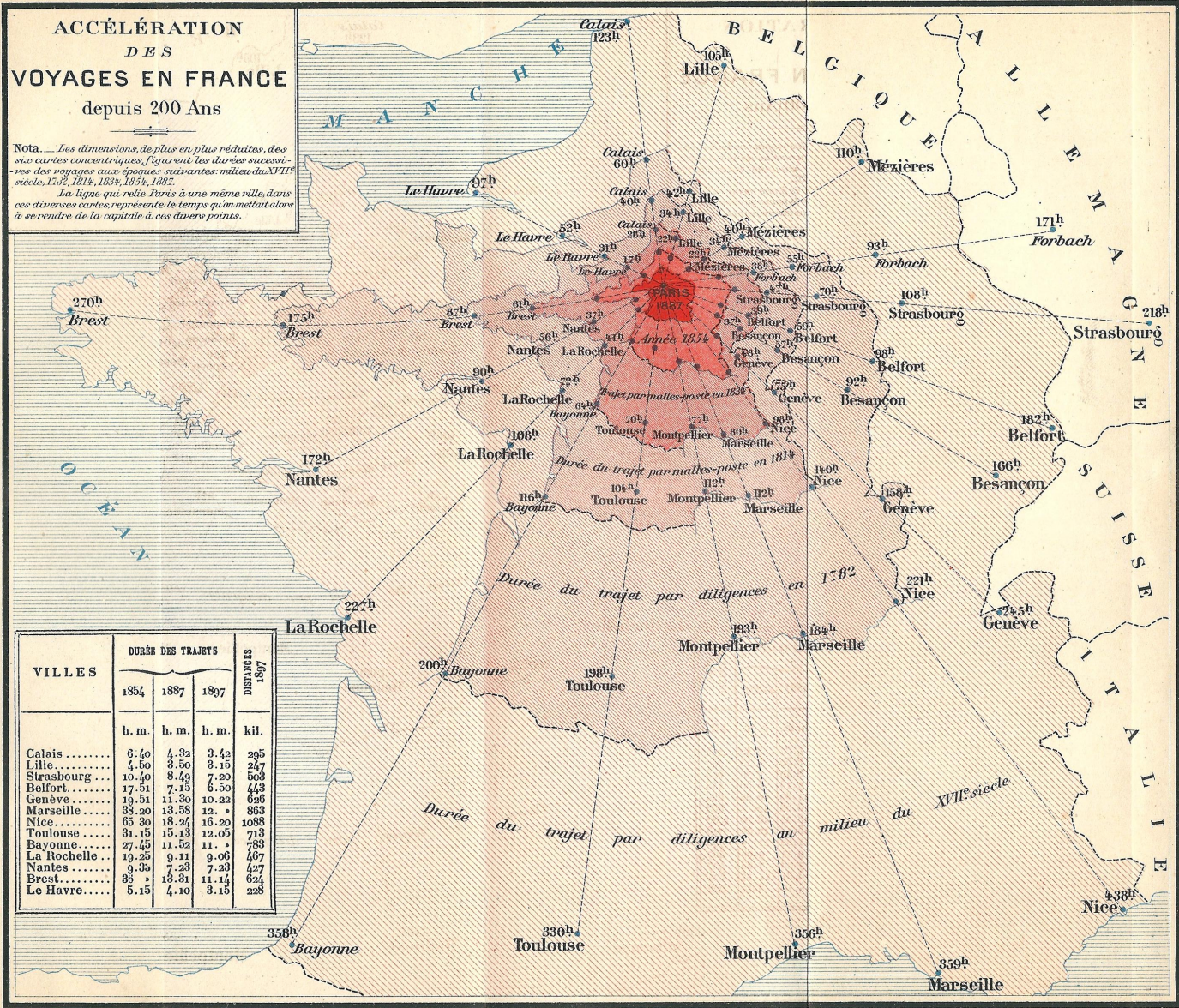


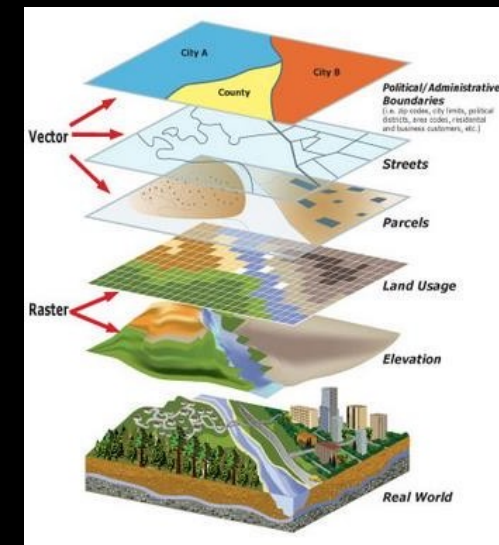
Map Spotlight 5: Thematic Map 2



Acceleration of Travel in France for 200 Years,
1888, Émile Cheysson

Category of Maps by Function

- General reference maps
 - General reference maps are to show the locations of a variety of different features, such as water bodies, coastlines, and roads.
- Thematic maps
 - Thematic maps or special purpose maps concentrates on **the distribution of a single attribute** or **the relationship among several**.
 - Basemap (底圖) + layers (圖層)
- Charts



Thematic Map (主題地圖)

- Thematic maps or special purpose maps concentrates on the distribution of a single attribute or the relationship among several.
- Thematic maps involve the display of **quantitative** data that are easily incorporated into a GIS or mapping software so that we can communicate the spatial variability of that data.
- The data are presented in a row and column format in which the geographic location, whether it be for a specific location that can occur at a point, along a line, or within a polygon, is normally represented in rows and the many attributes collected for that location are presented in the columns.
- We investigate the **statistical characteristics** of the data and how they assist us in the selection of which thematic map to use.

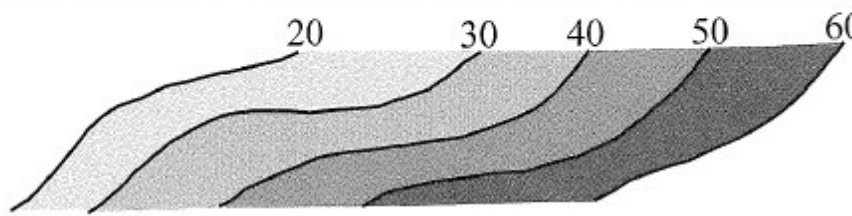
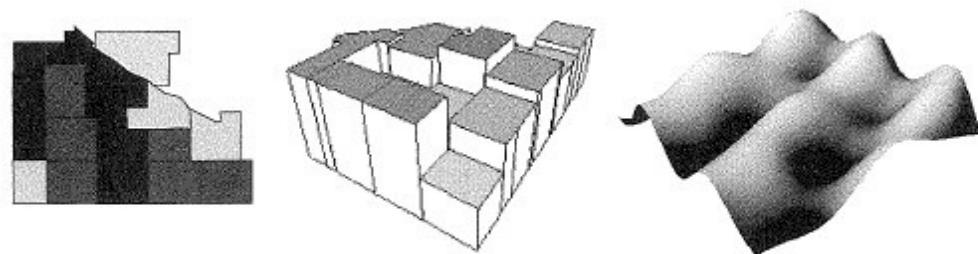
Types of Thematic Maps

- Point maps (點位地圖)
- Line maps (線狀地圖)
- Area maps (面狀地圖)
- Cartograms (示意地圖)
- Chartmaps (圖表地圖)
- 3D maps (三維地圖)

Area Maps

- Areas can vary in size and scale from very small (precinct or district) to very large (state or even country).
- Population data usually is reported in aggregate form using administrative areas.
- Choropleth maps (面量圖 ; 等值區域圖) are a common technique for representing data for areas by variations in shading or pattern.
- Dasymetric maps (分區比重圖) are based on the principle that areas used to collect and report population data (e.g. counties) do not have a homogenous population that exhausts space.

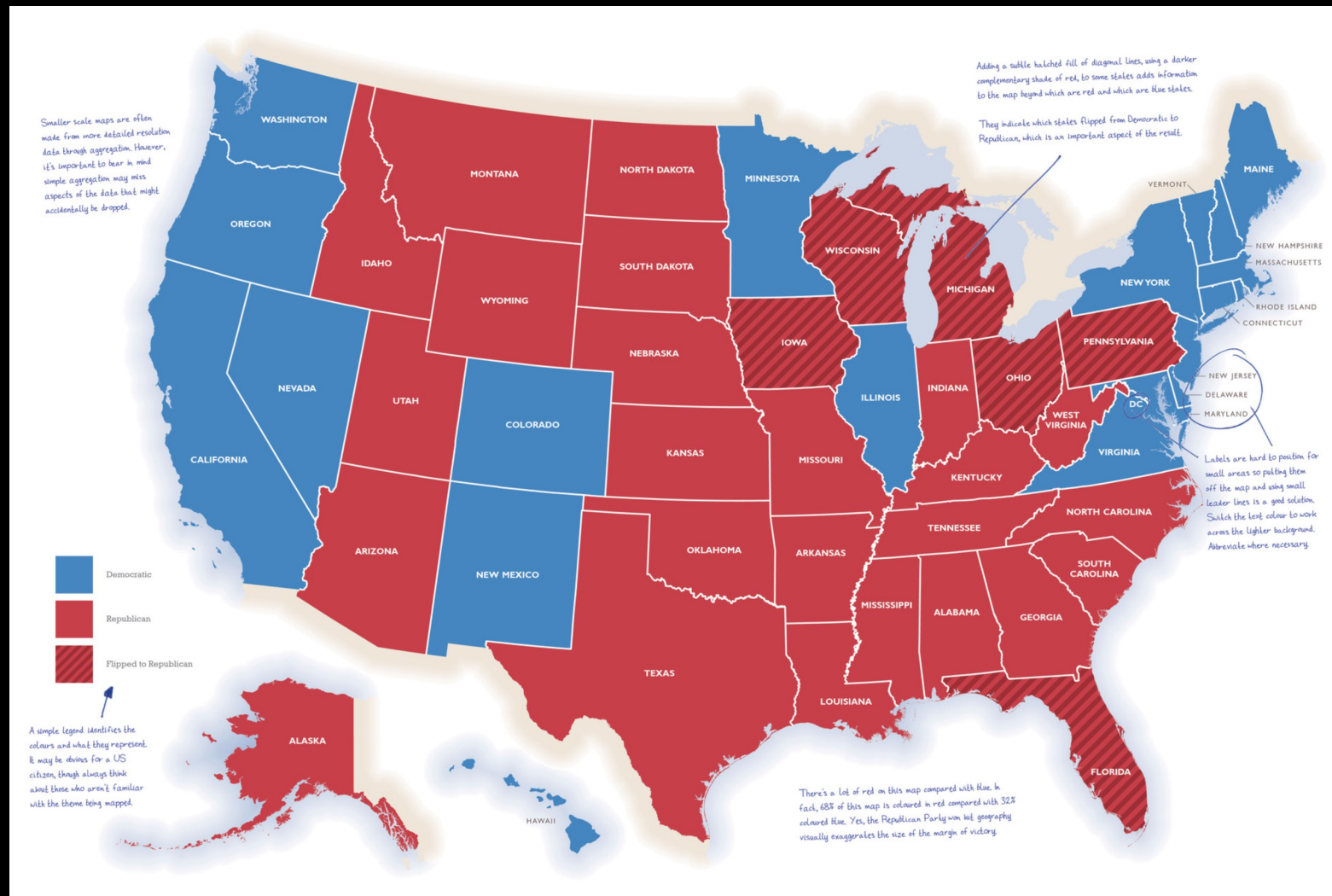
Area Symbols

Qualitative By Location and Nature		Quantitative By Location and Amount	
Area Symbols			
 Lake  Marsh  Sand  Forest		Bounded By Isolines	
		Bounded By Non-Isolines	



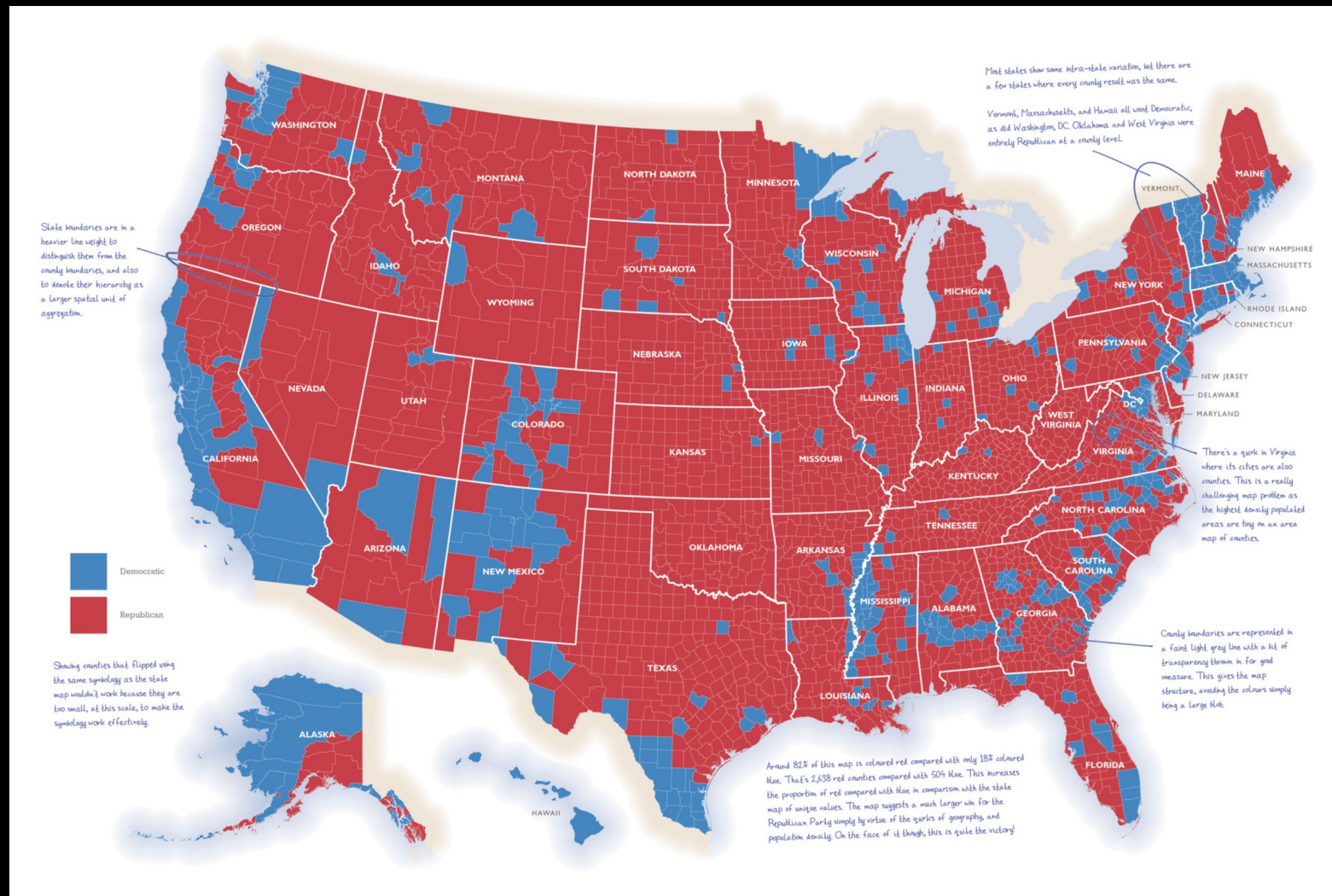
First dasymetric map (分區比重圖): Population density, 1833, George Julius Poulett Scrope

The bottom-line result as unique values



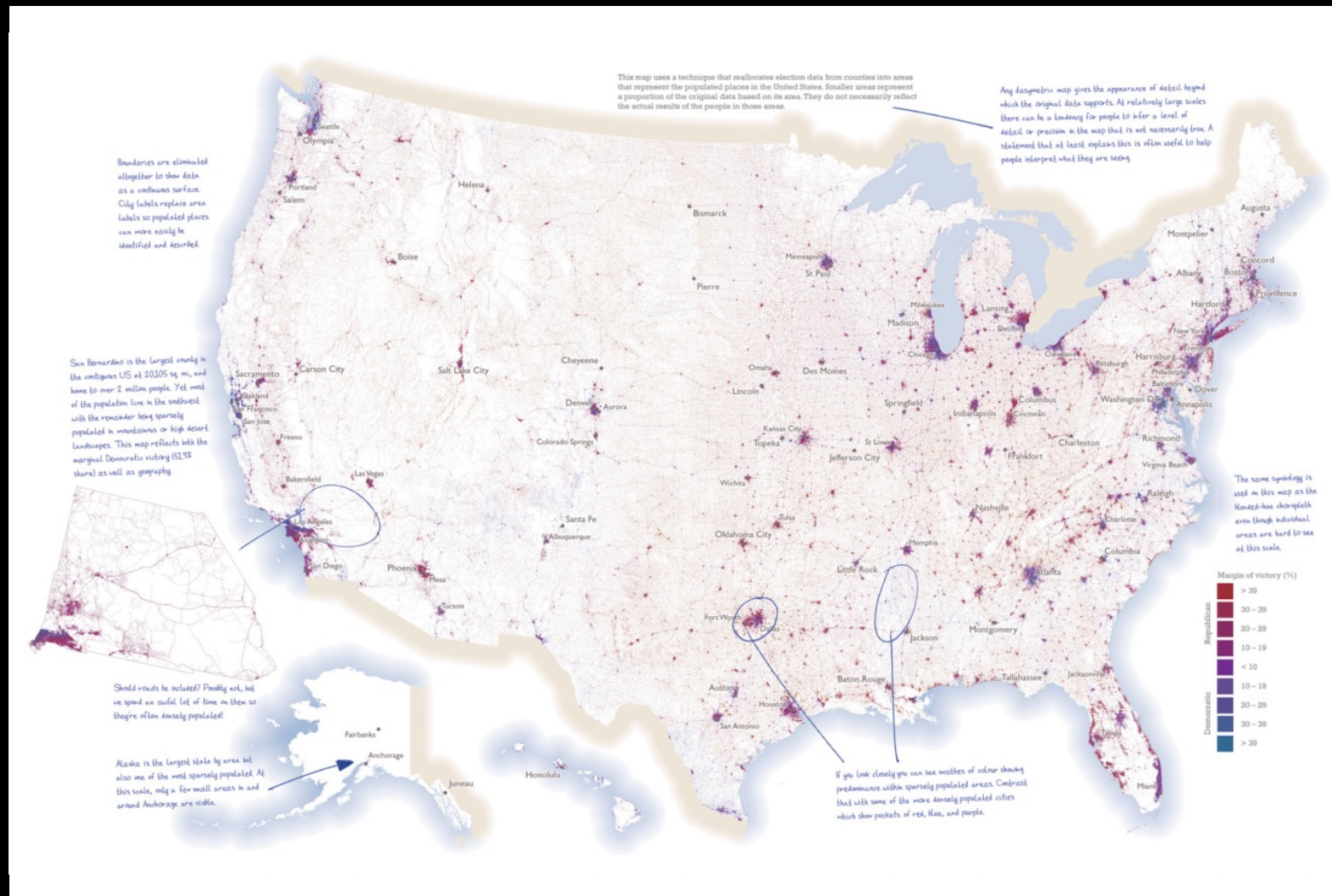
Field (2022)

Unique values at a finer level of details



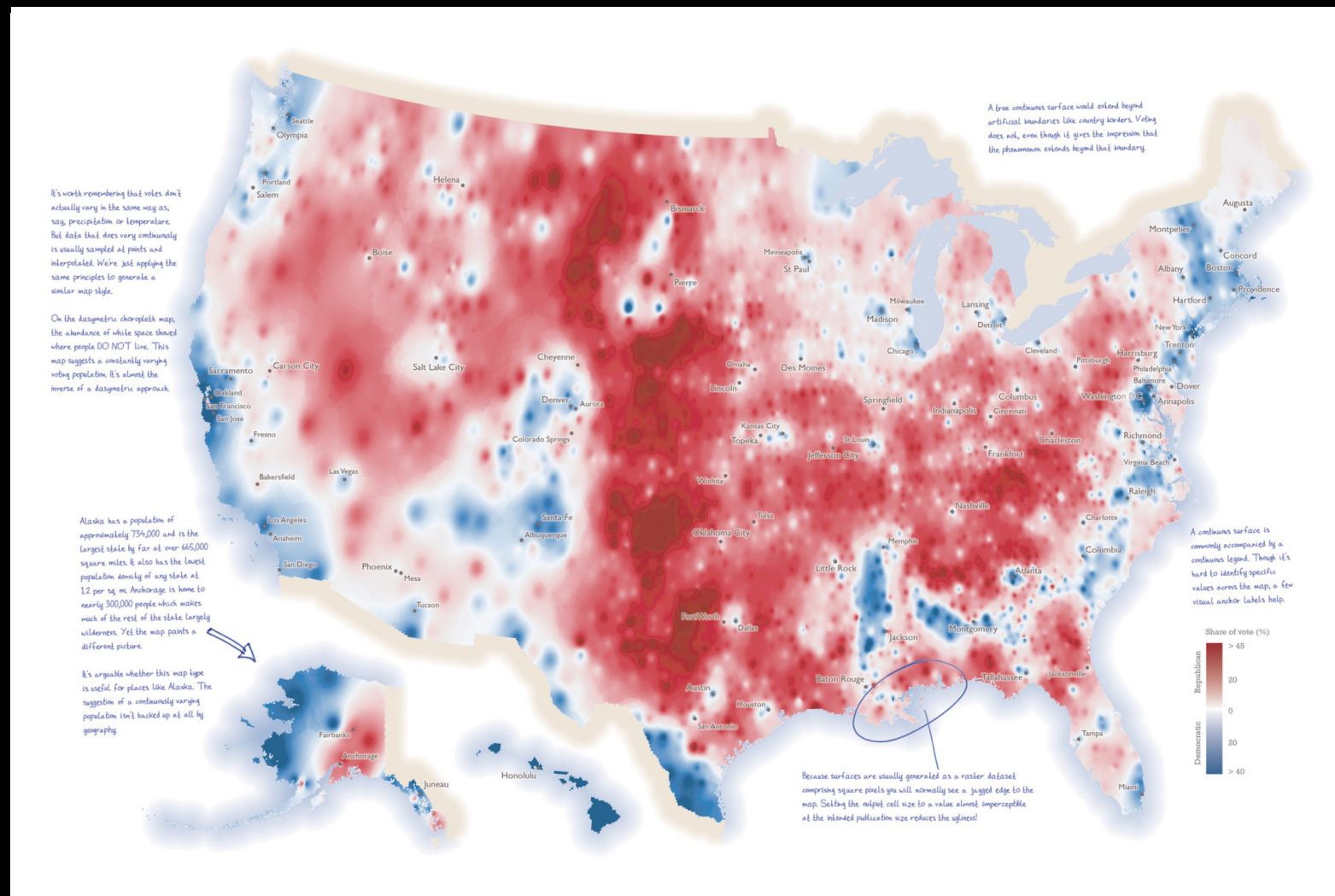
Field (2022)

Dasymetric reallocation to new areas



Field (2022)

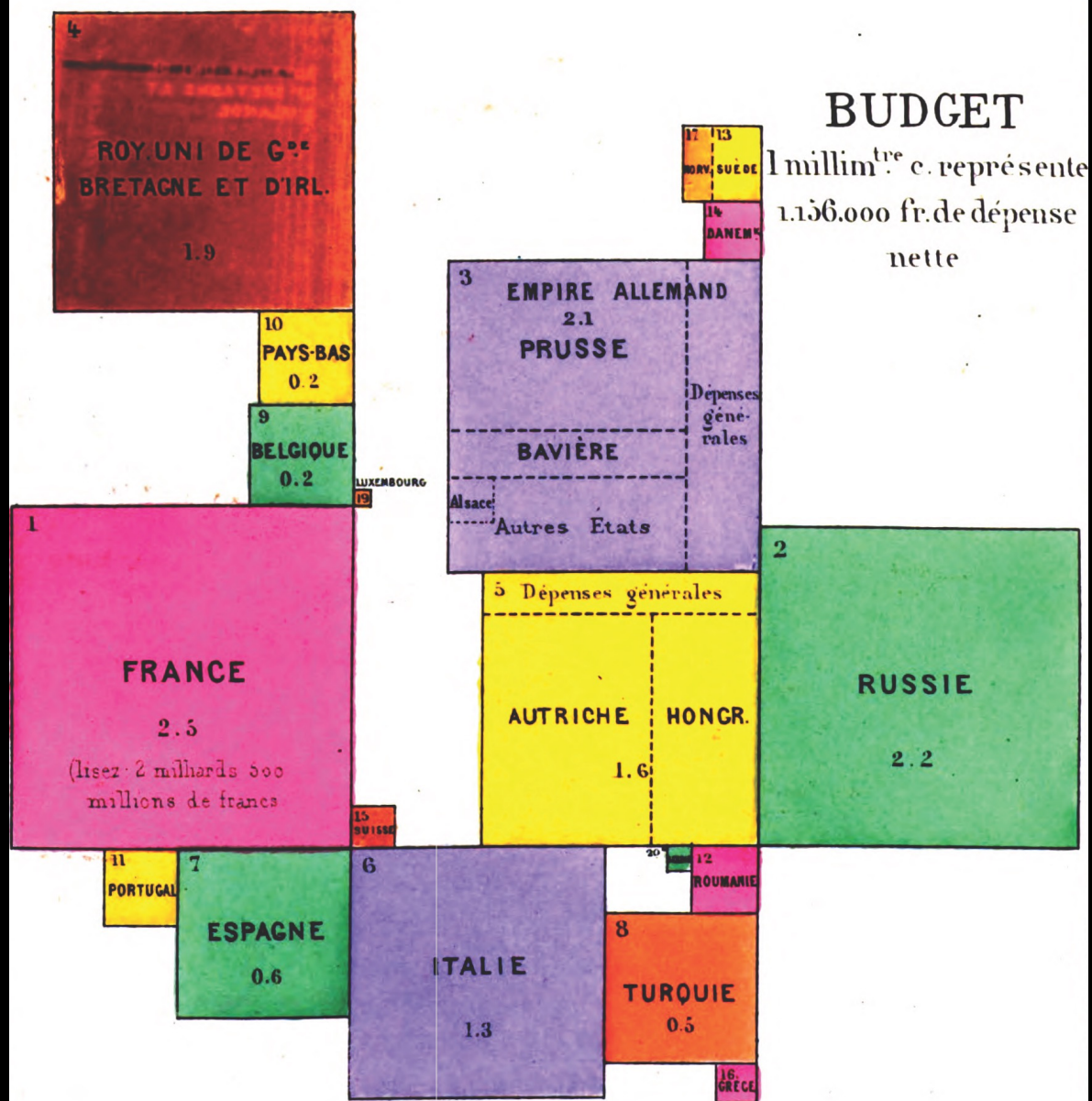
Interpolated surfaces (Continuously varying results)



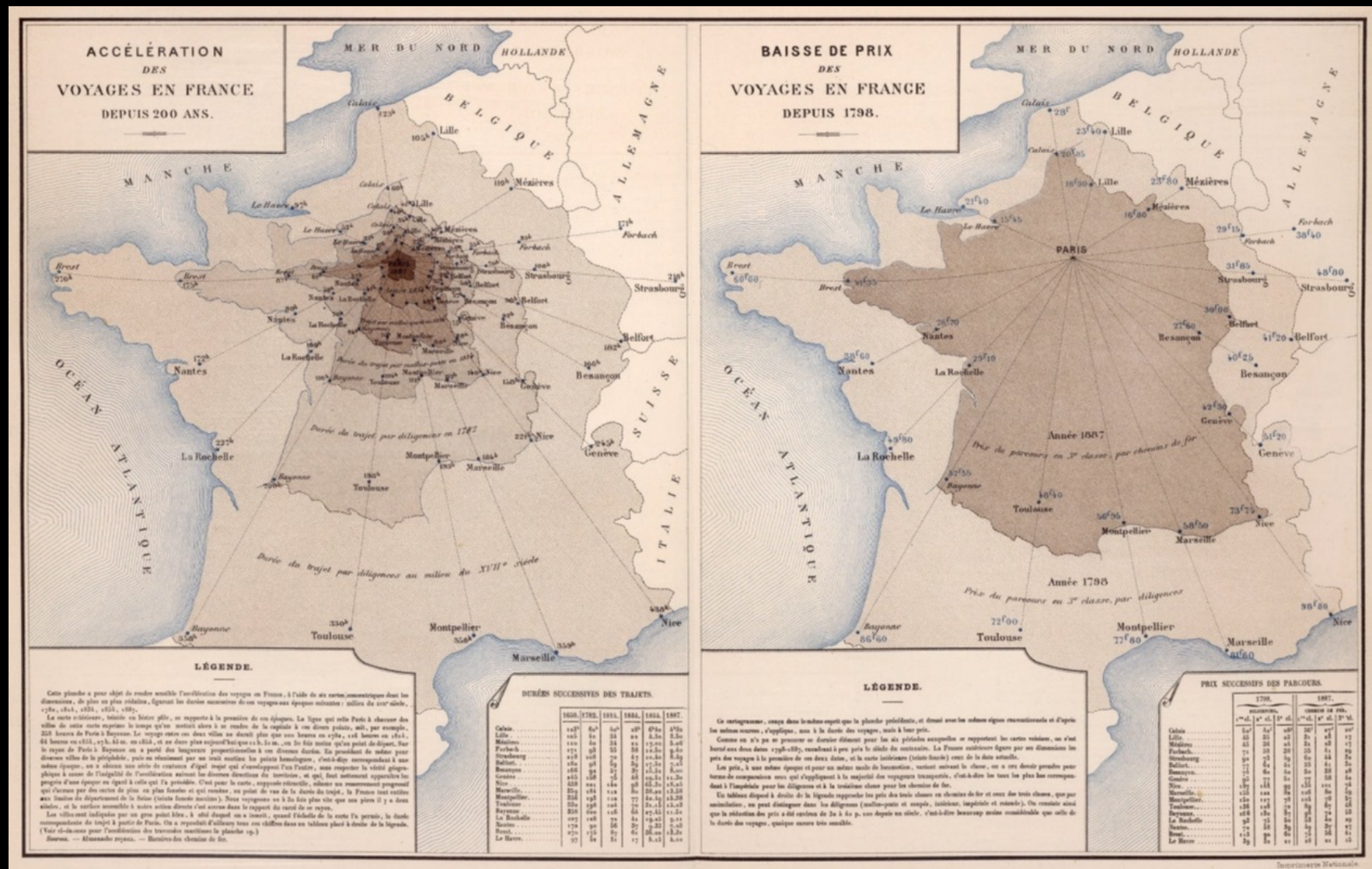
Field (2022)

Cartograms (示意地圖)

- Cartograms are to reimagine or warp it, to nullify the impact of geography's complexity on the core message of the map.
- Cartograms tackle the inherent visual biases in geographical maps by modifying the geometry of the geography, effectively normalizing it to create a map in which each area takes on a new shape or size.
- Cartograms therefore depict geographical space diagrammatically as they lose their relationship with actual geographic coordinates.
- One disadvantage is that they change the visual representation of geography.

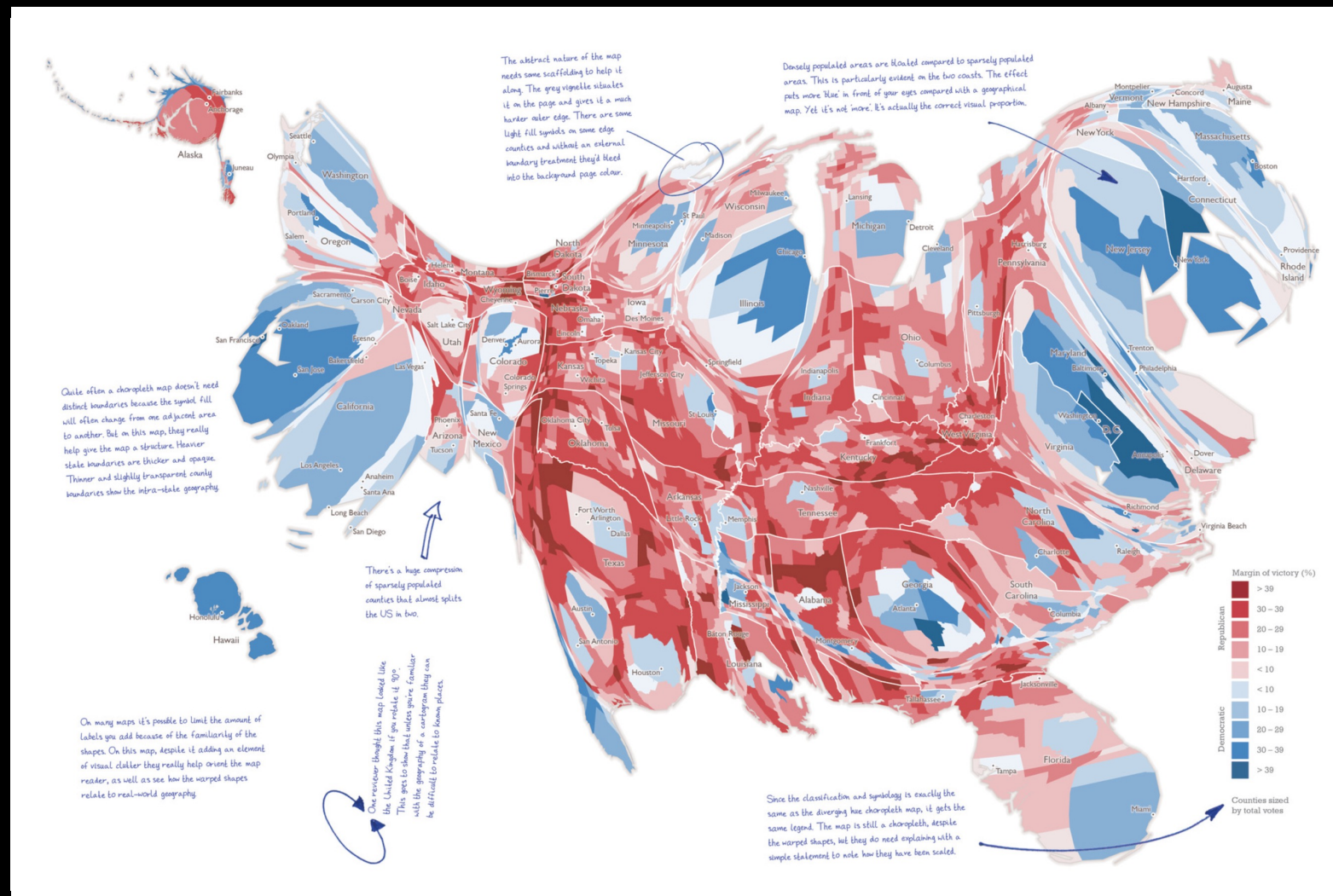


First cartogram:
International Statistical Congress in Budapest,
1876, Pierre Émile Levasseur



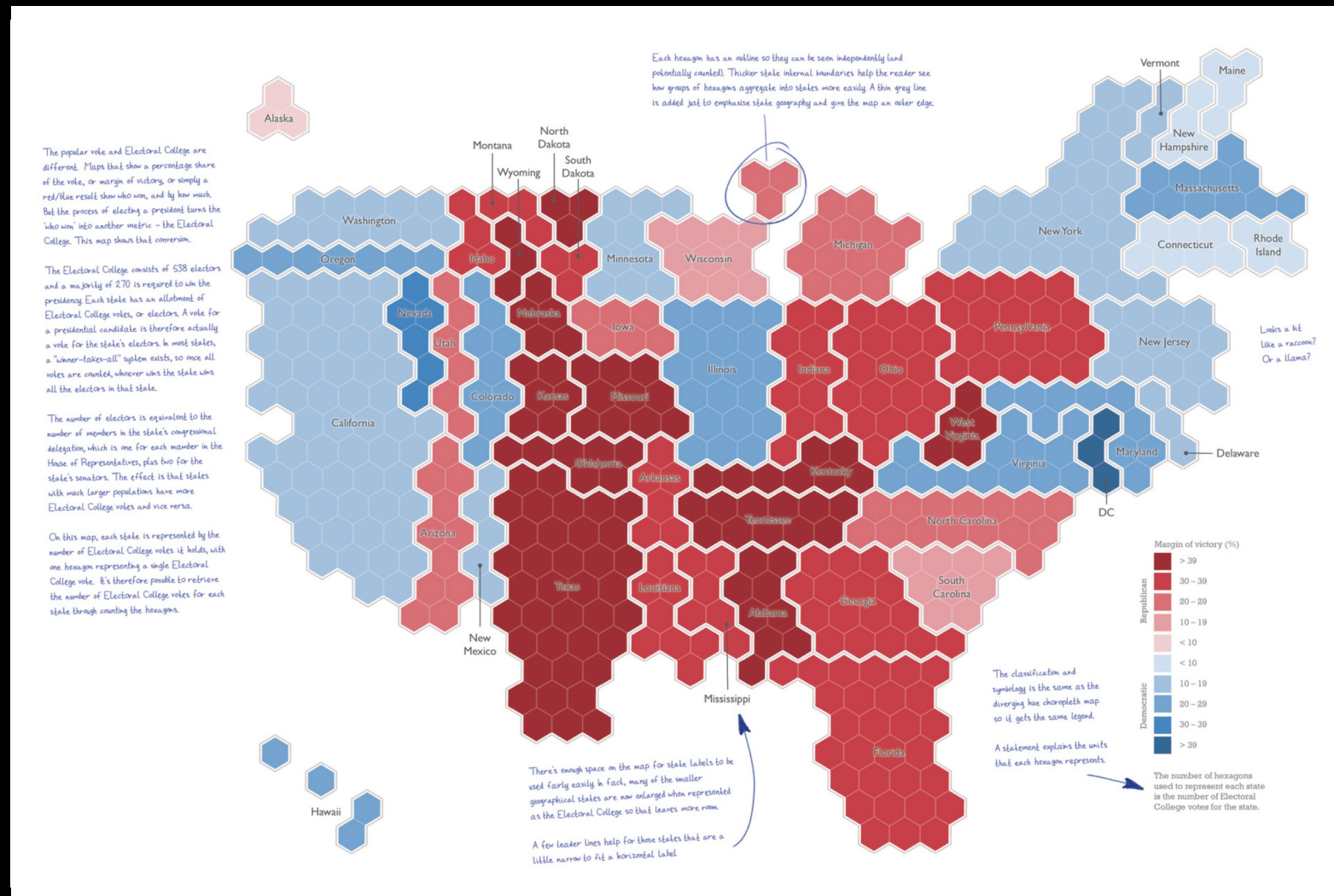
First anamorphic map (變形地圖): Acceleration of Travel in France for 200 Years, 1888, Émile Cheysson

Population density-equalizing cartogram (Out of shape)



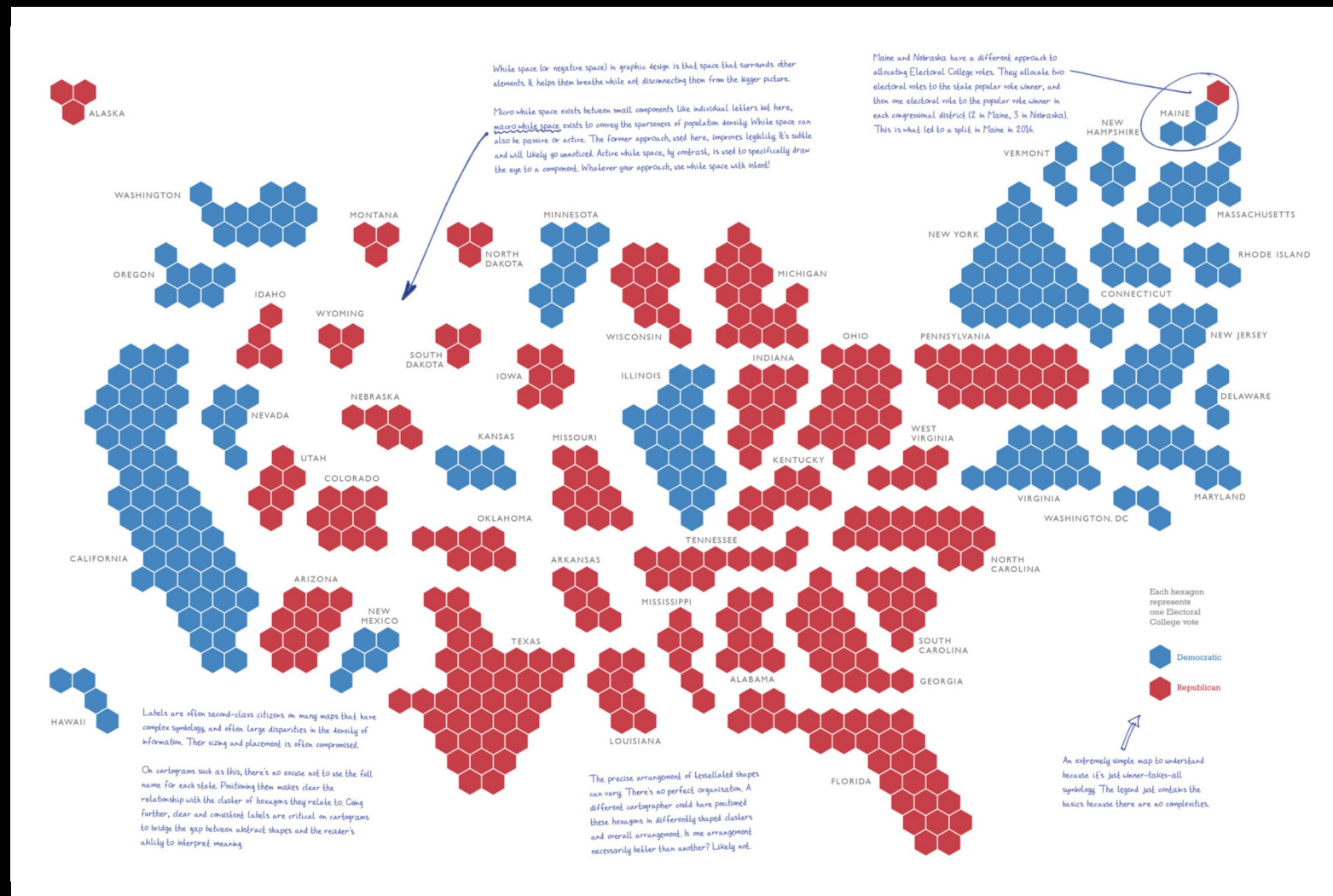
Field (2022)

Gridded cartogram by hexagon



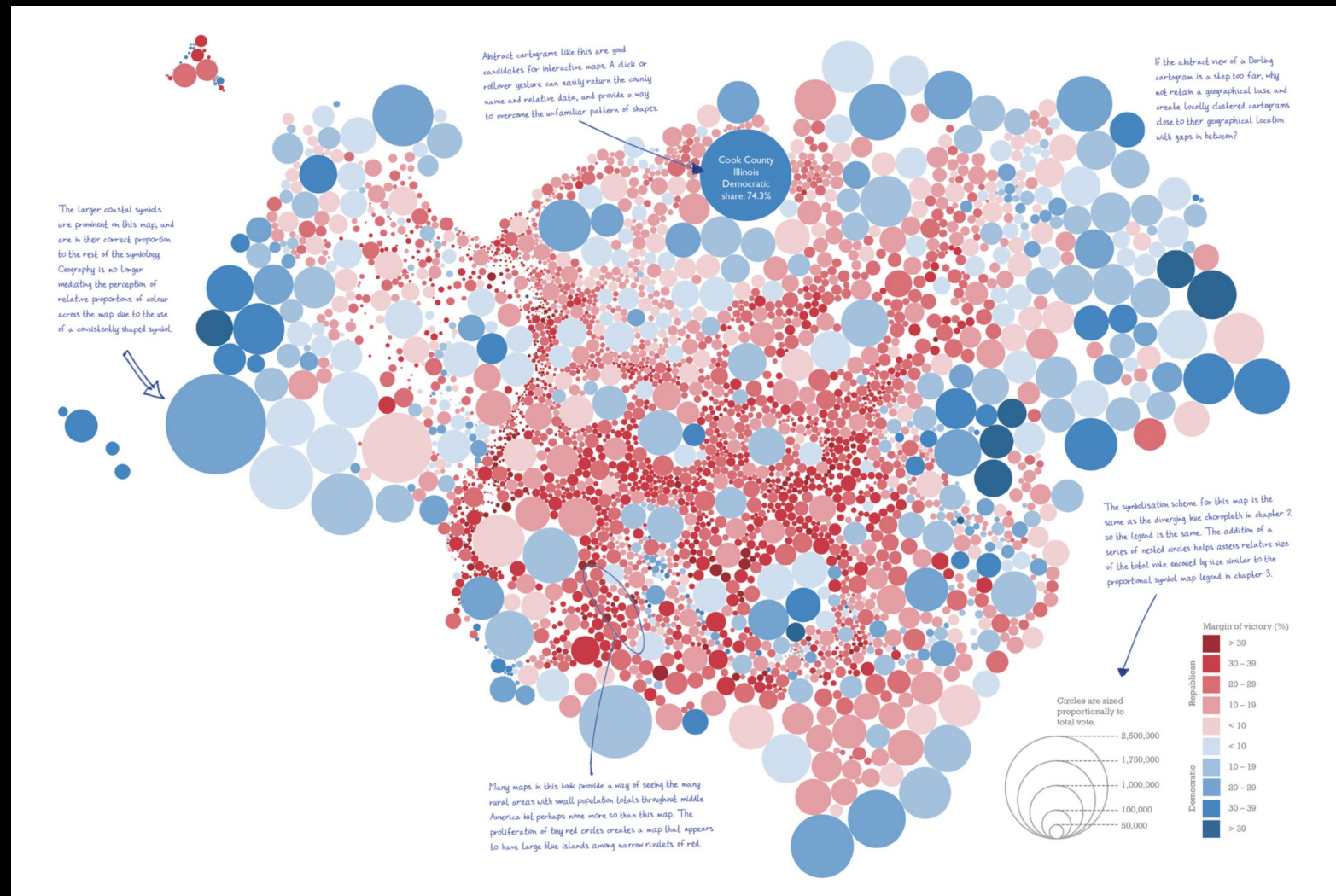
Field (2022)

Non-contiguous gridded cartogram



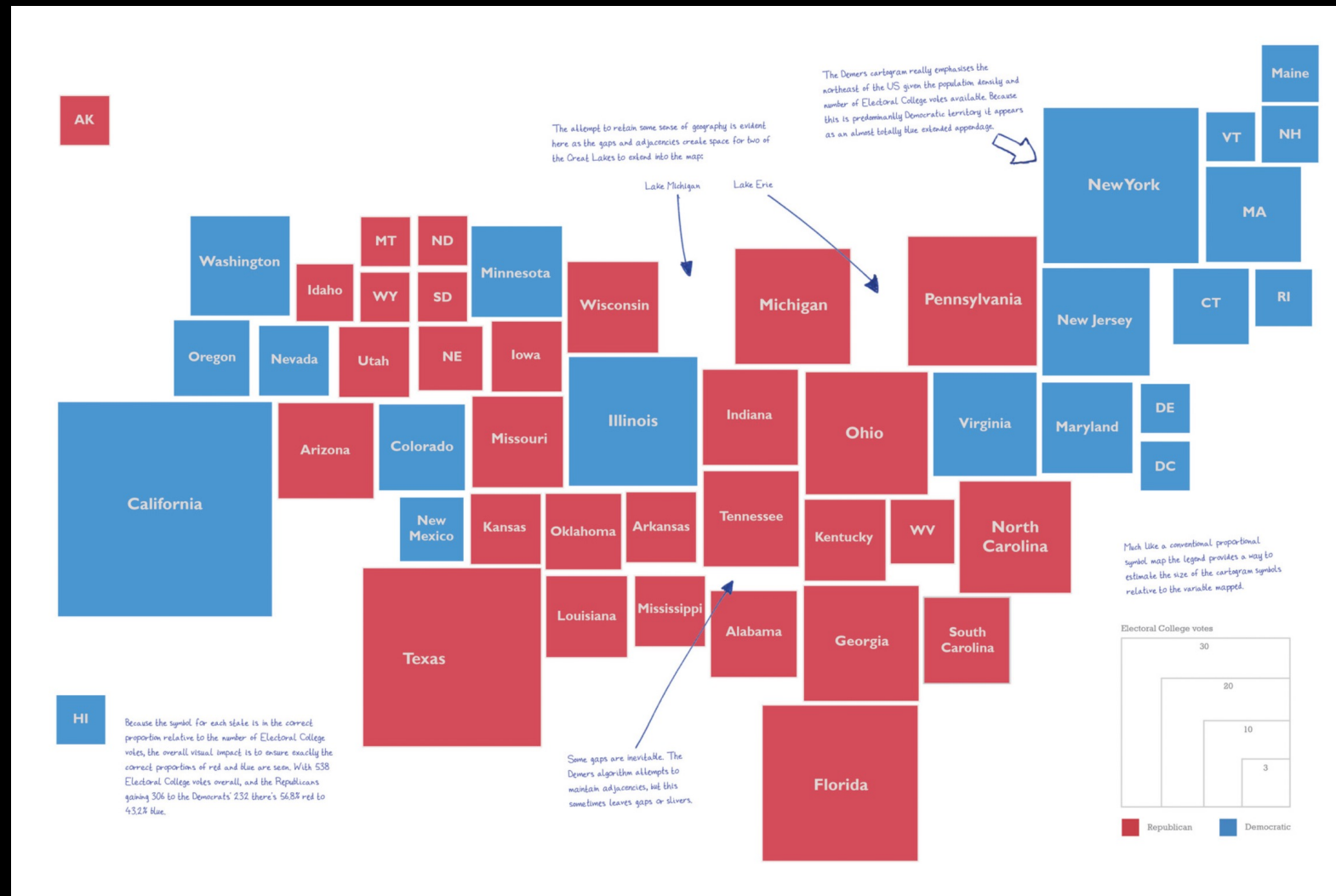
Field (2022)

Dorling cartogram



Field (2022)

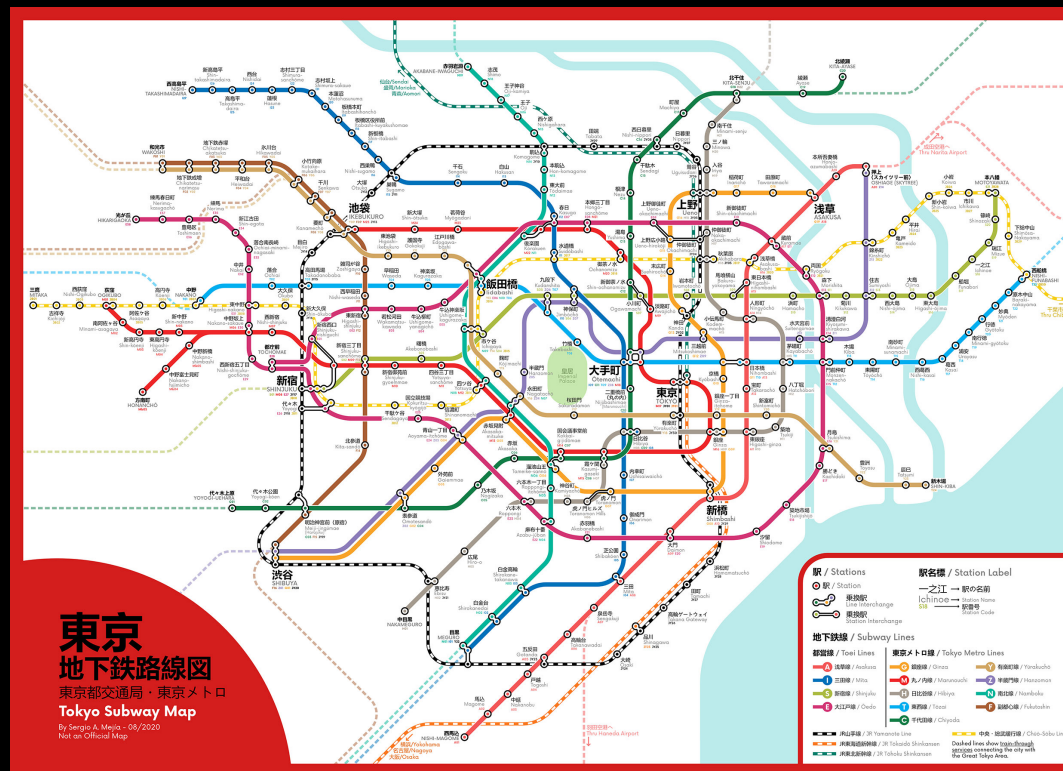
Demers cartogram



Field (2022)

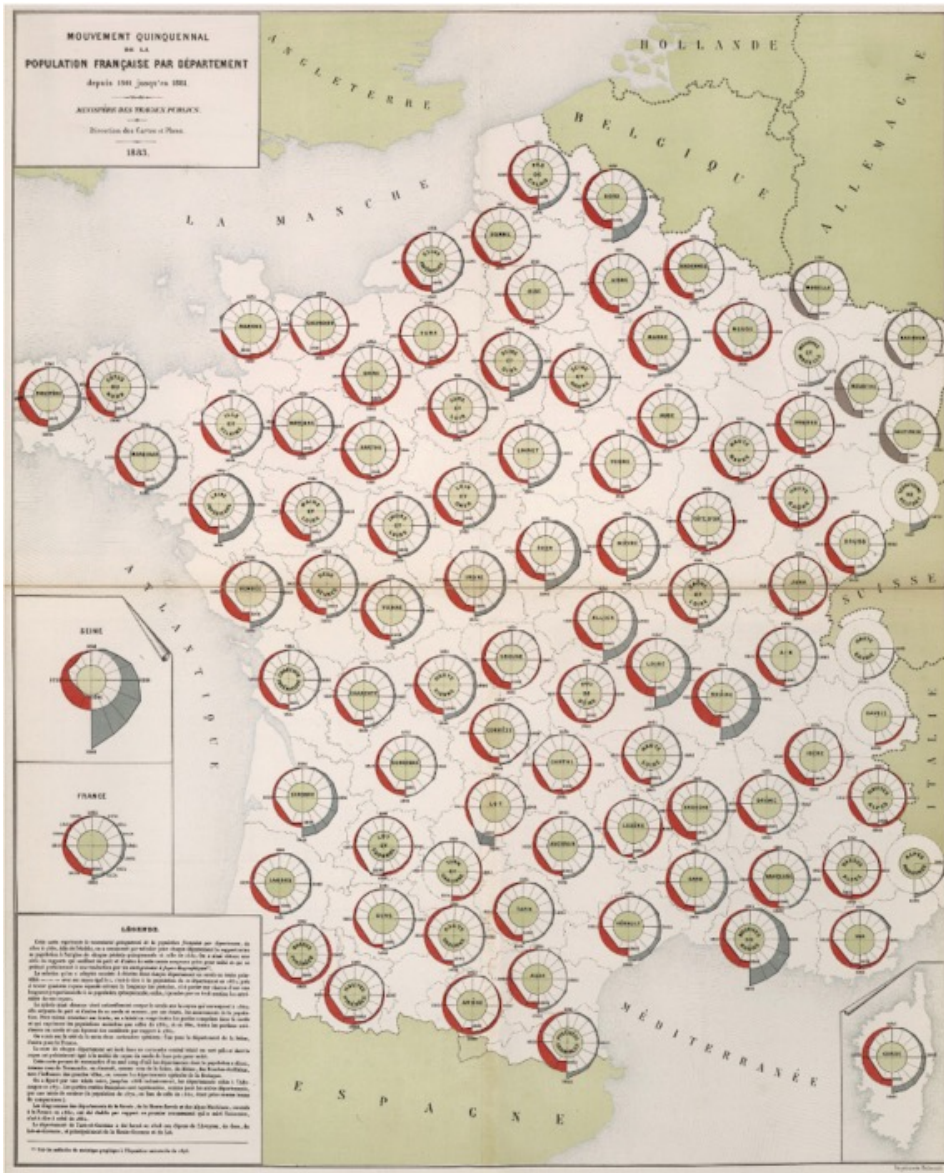
Topological Map (拓樸地圖)

- Topological maps are a type of diagram only concerned with relative locations of features on the map.
- Don't be confused with topographical maps (地形圖).

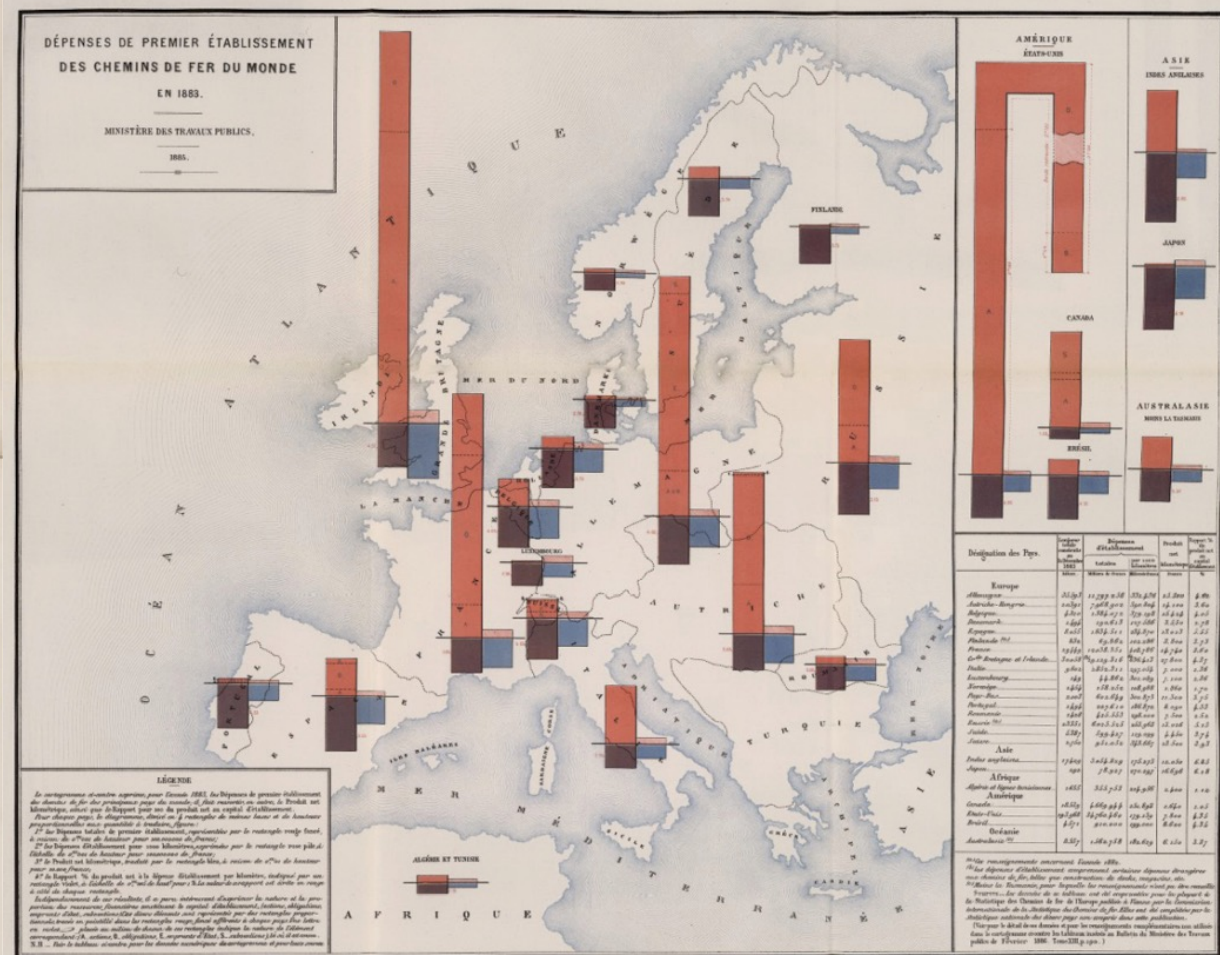


Chartmaps (圖表地圖)

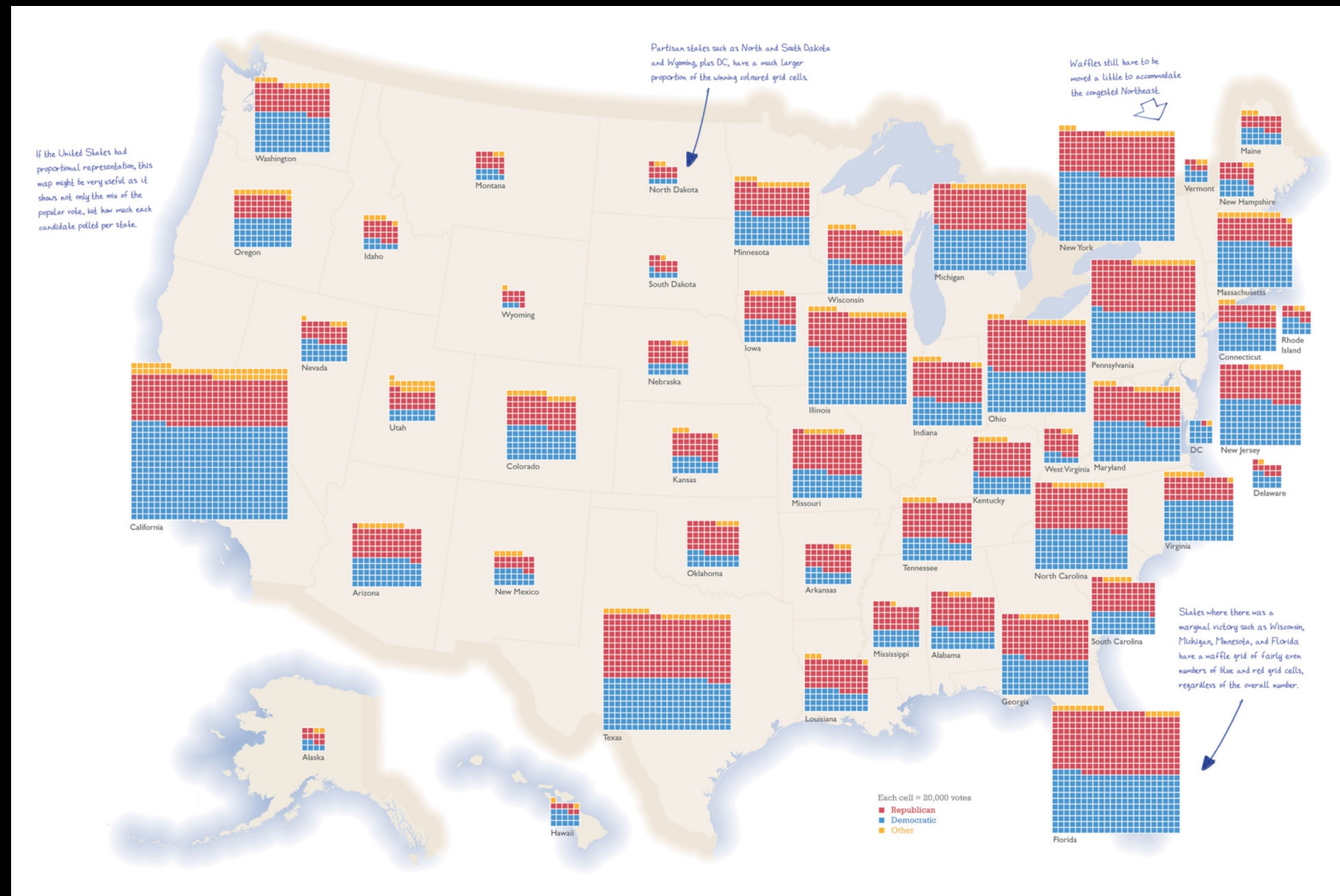
- A set of useful visual tools allows the presentation of data that teases out important components – namely, graphs, charts, and plots.
- Put graphs and maps together, and it creates a powerful mechanism to explore not only patterns in the data (via the graphs), but also how those patterns exist in different places.
- Chartmaps often resort to abstract geographies. Some use a geographical background whereas others use a gridded cartogram to provide consistently sized graphical containers.



Population growth in France, 1883, Émile Cheysson

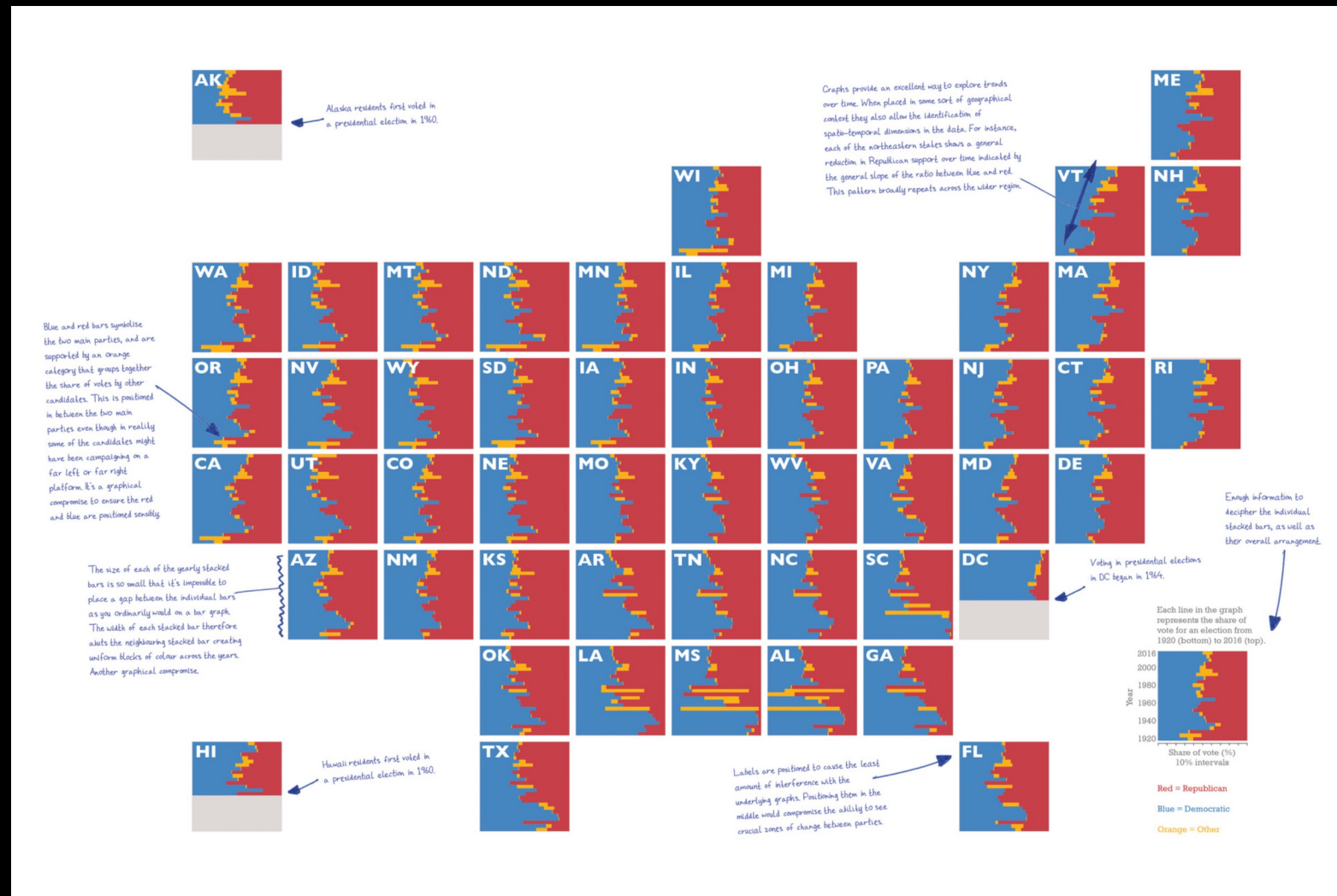


Waffle grid on a geographical base



Field (2022)

Small multiples stacked bar charts



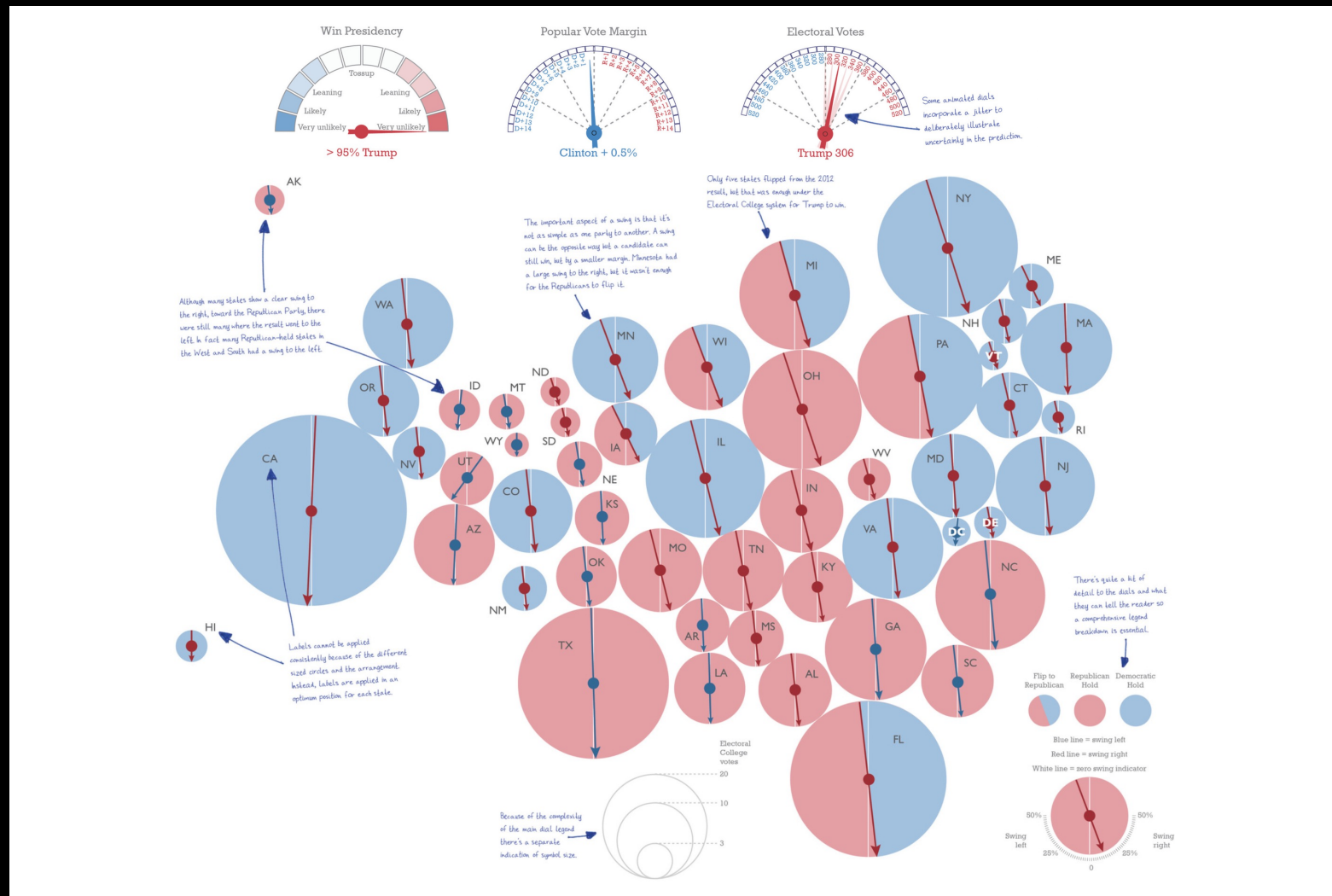
Field (2022)

Minimalist sparklines



Field (2022)

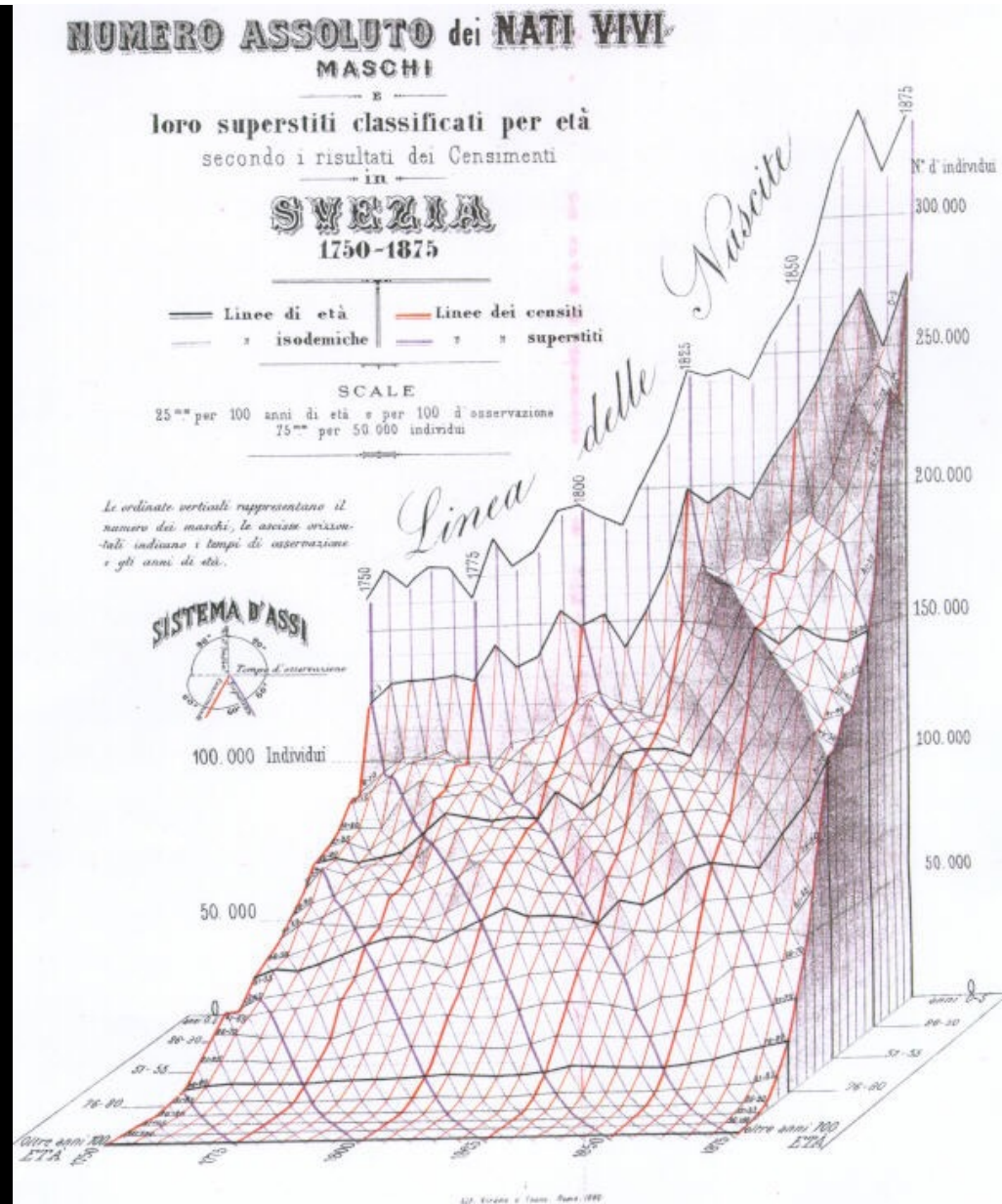
Dials as a Dorling cartogram



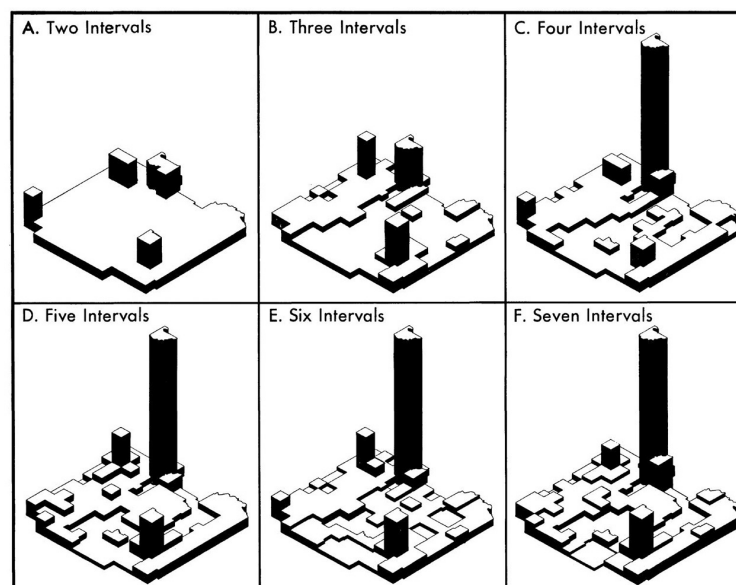
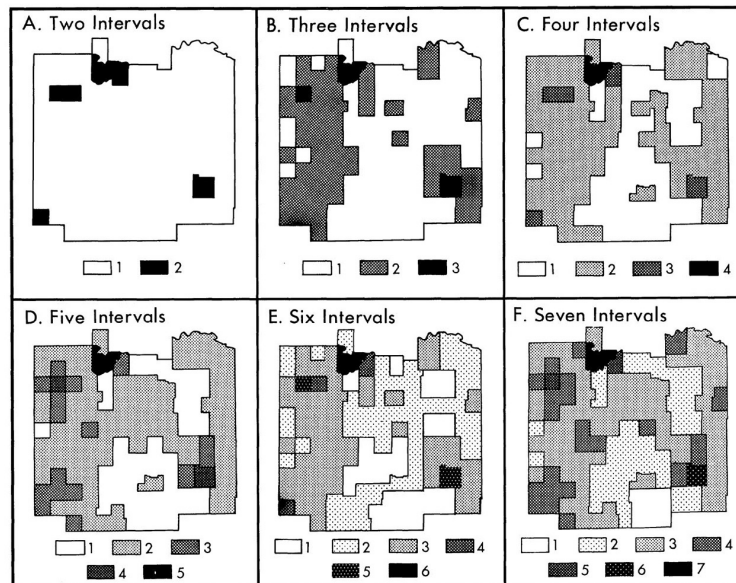
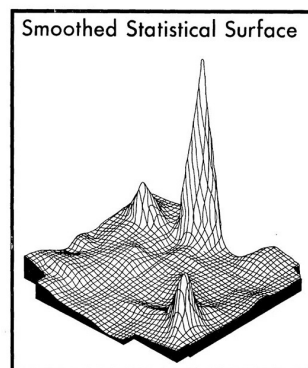
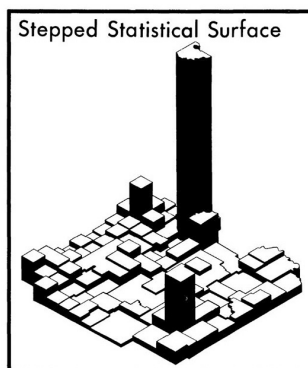
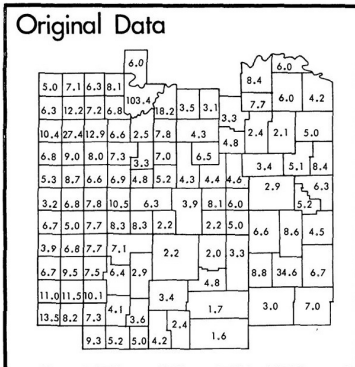
Field (2022)

3D Maps

- 3D maps have become more popular as a vehicle to display data, particularly on the web.
- Human naturally see the world in three dimensions via perspective views so it makes sense to think that data can be presented in a similar way.
- The key to an effective 3D map is to determine when it is appropriate to take advantage of the z-dimension to bring something meaningful to the map.
- It requires careful handling statistically or via an interactive web environment.



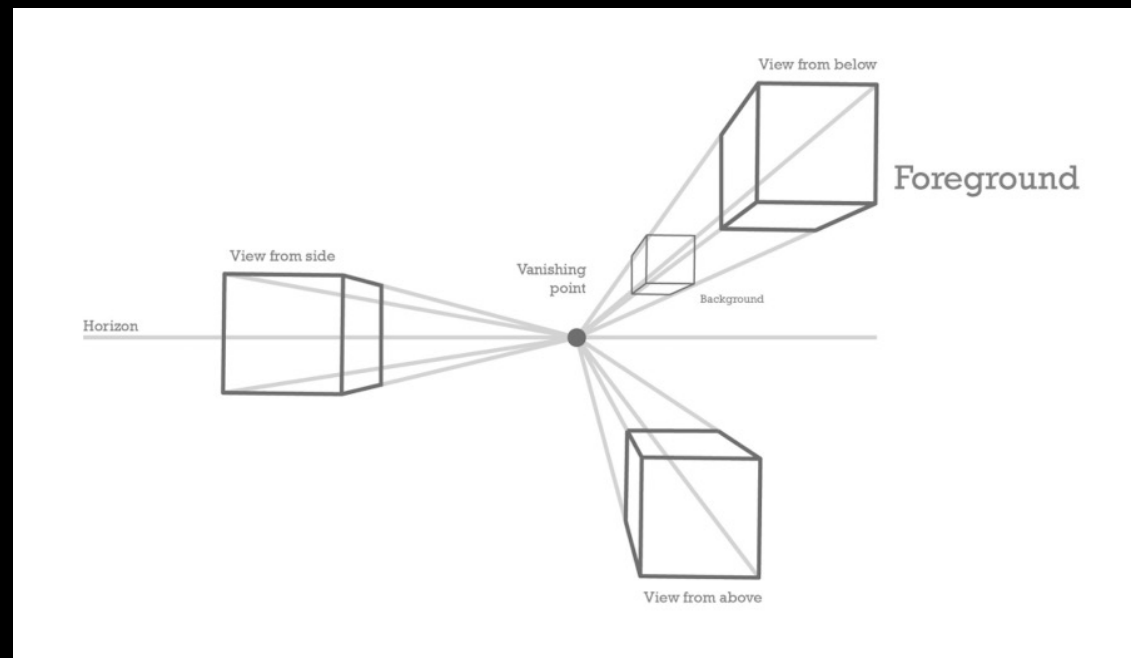
First block diagram
3D population pyramid of live male births,
1879, Luigi Perozzo



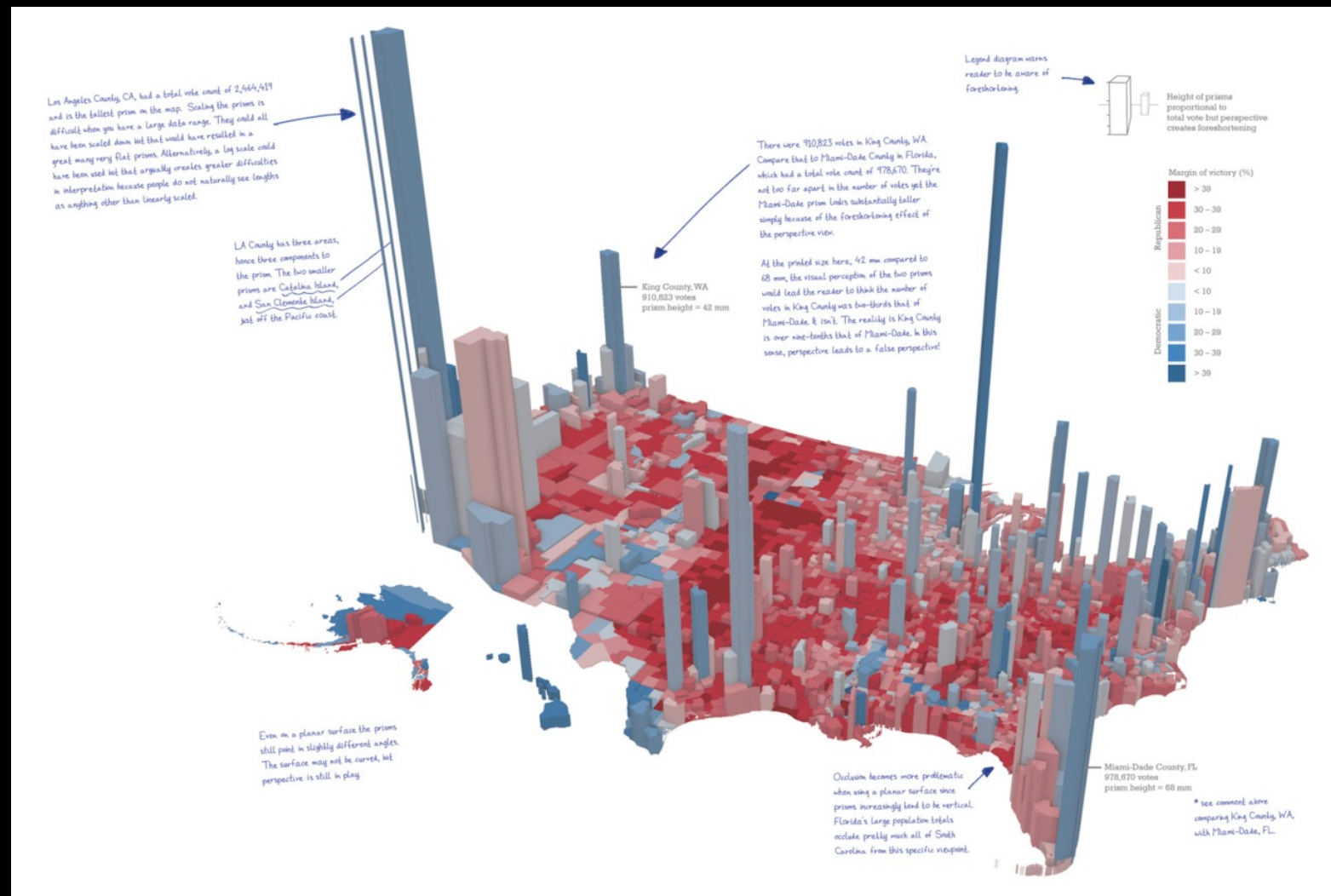
First 3D prism map (稜柱圖):
Stepped Statistical Surface, 1963, George Jenks

3D View

- Conventionally, 3D models of data are viewed from a single viewpoint (單一視角) and often in perspective (透視).
- Perspective refers to the perception of depth in the image and is a function of where the viewpoint is set.



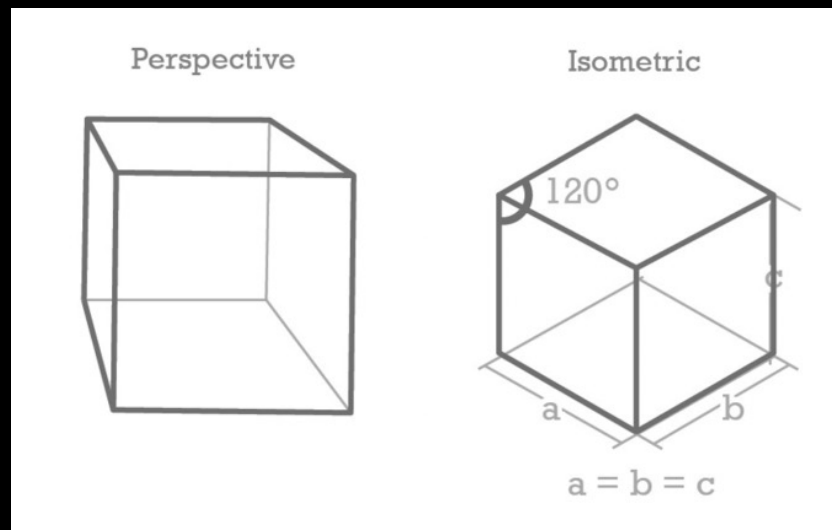
Extruded prisms in perspective



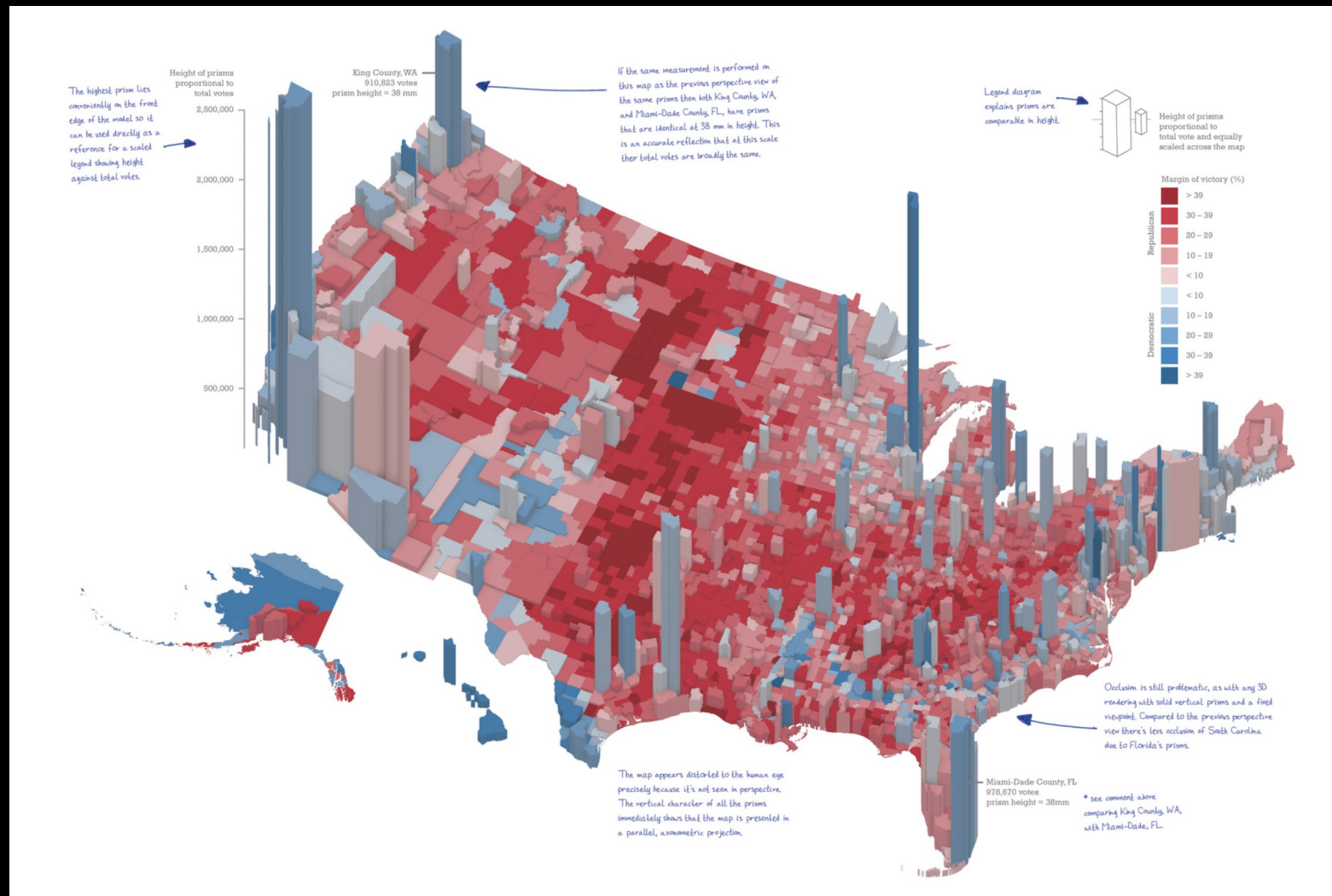
Field (2022)

3D View

- However, objects in the foreground will appear much larger than if they are set in the distance.
- Axonometric view (軸測投影) is a form of parallel, or orthographic projection in which sight lines are perpendicular to the plane of projection.
- All objects of the 3D model are equally foreshortened, meaning the scale is consistent across the entire map.

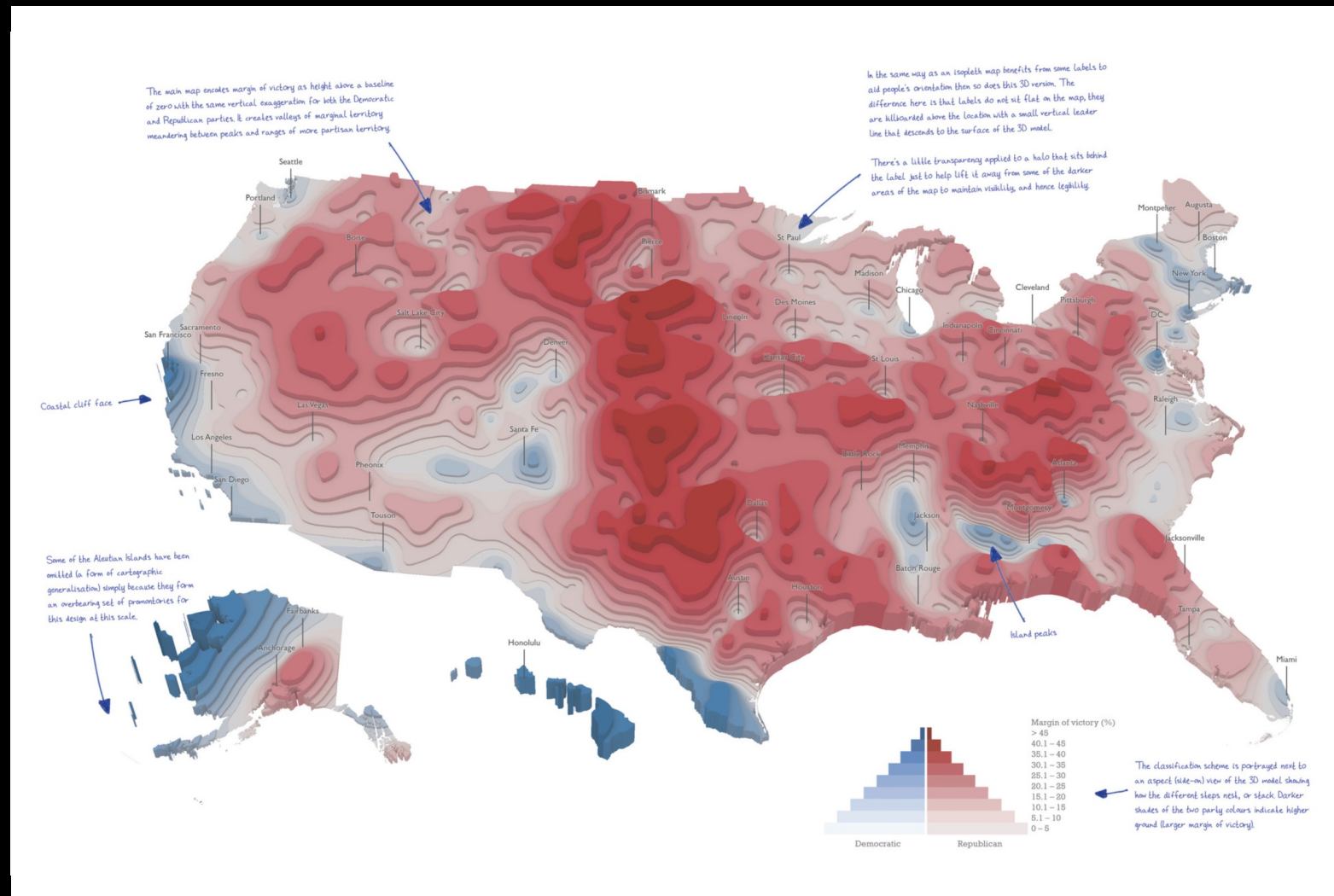


Extruded prisms in an axonometric view



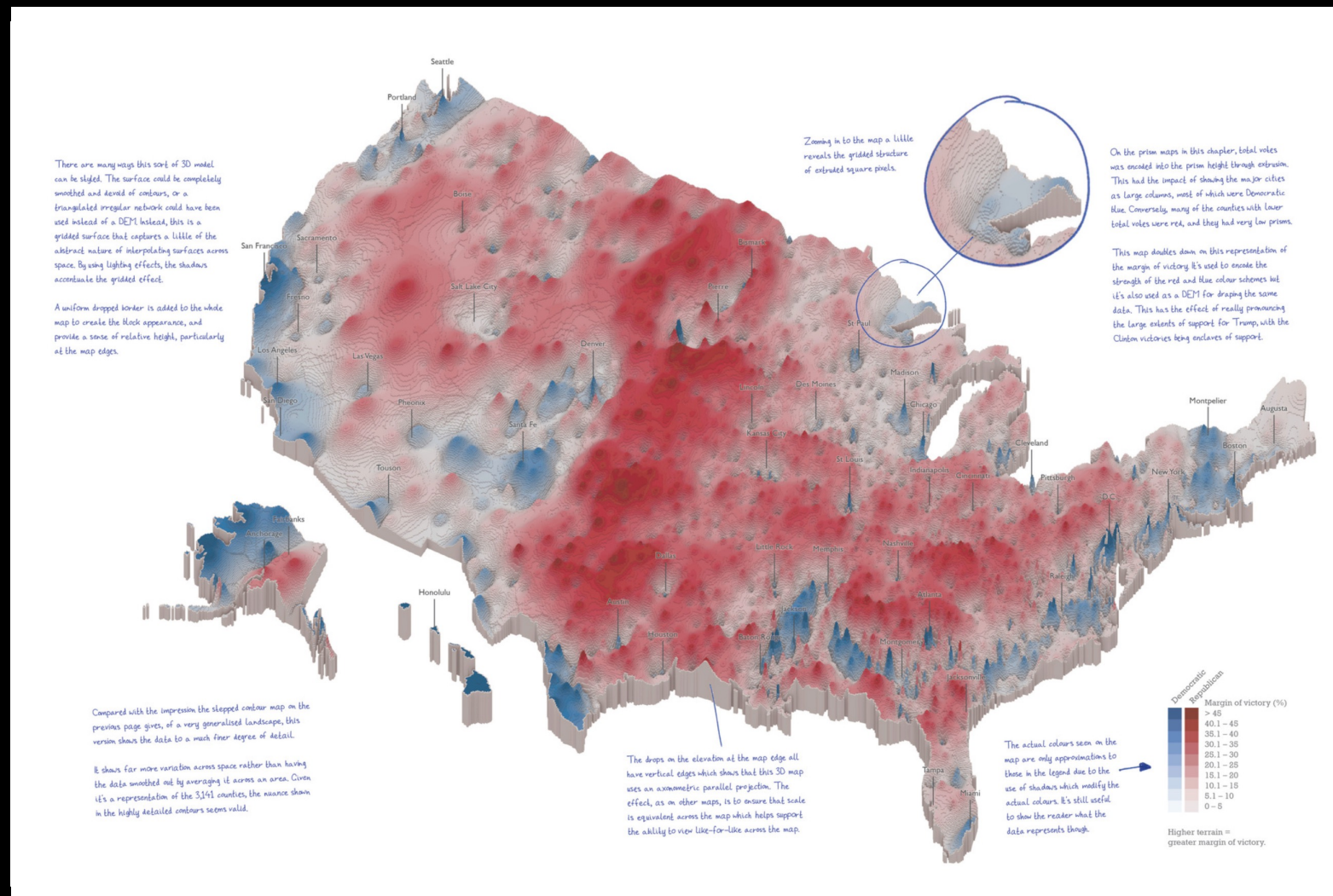
Field (2022)

Extruded filled contours



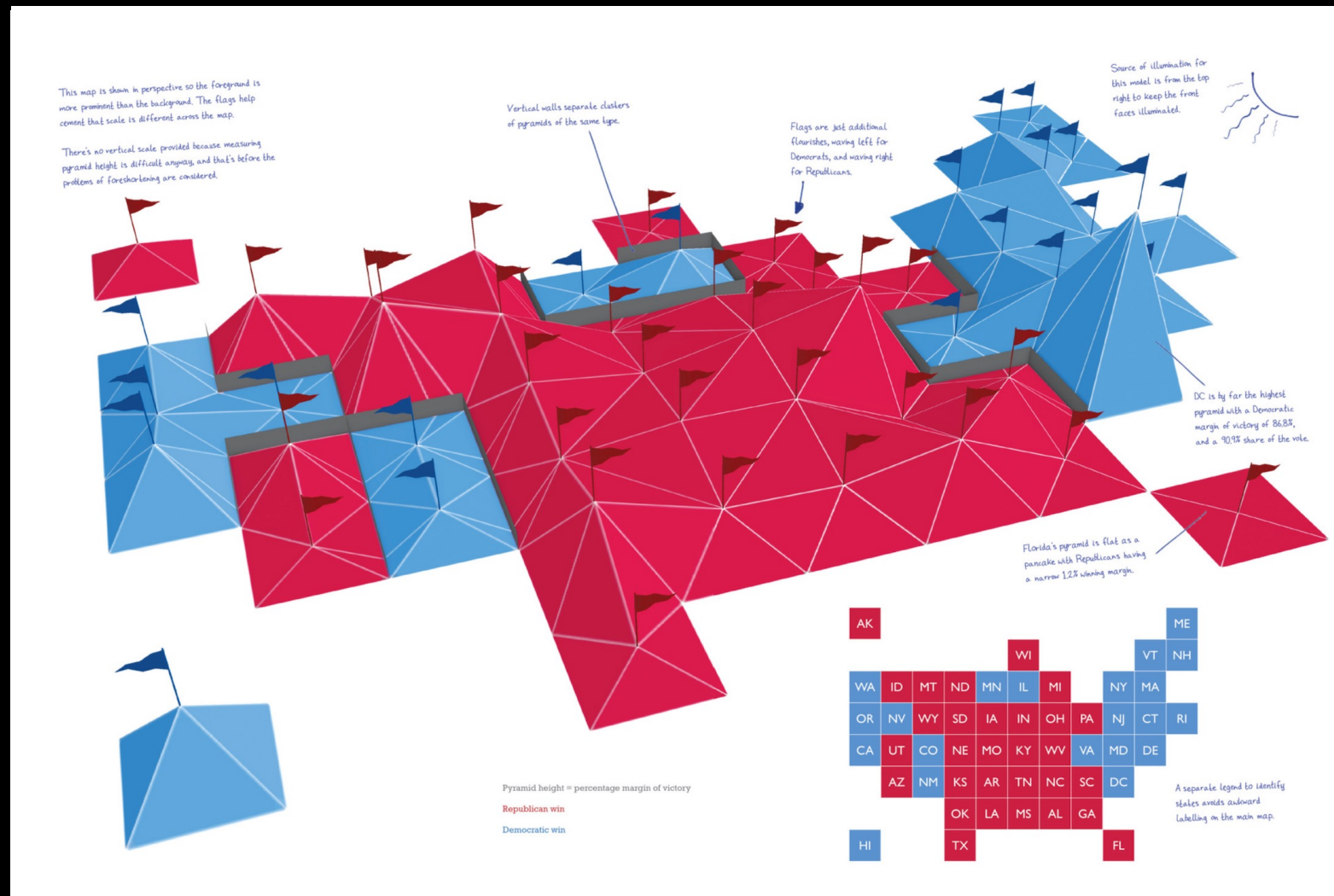
Field (2022)

3D block diagram with a draped surface



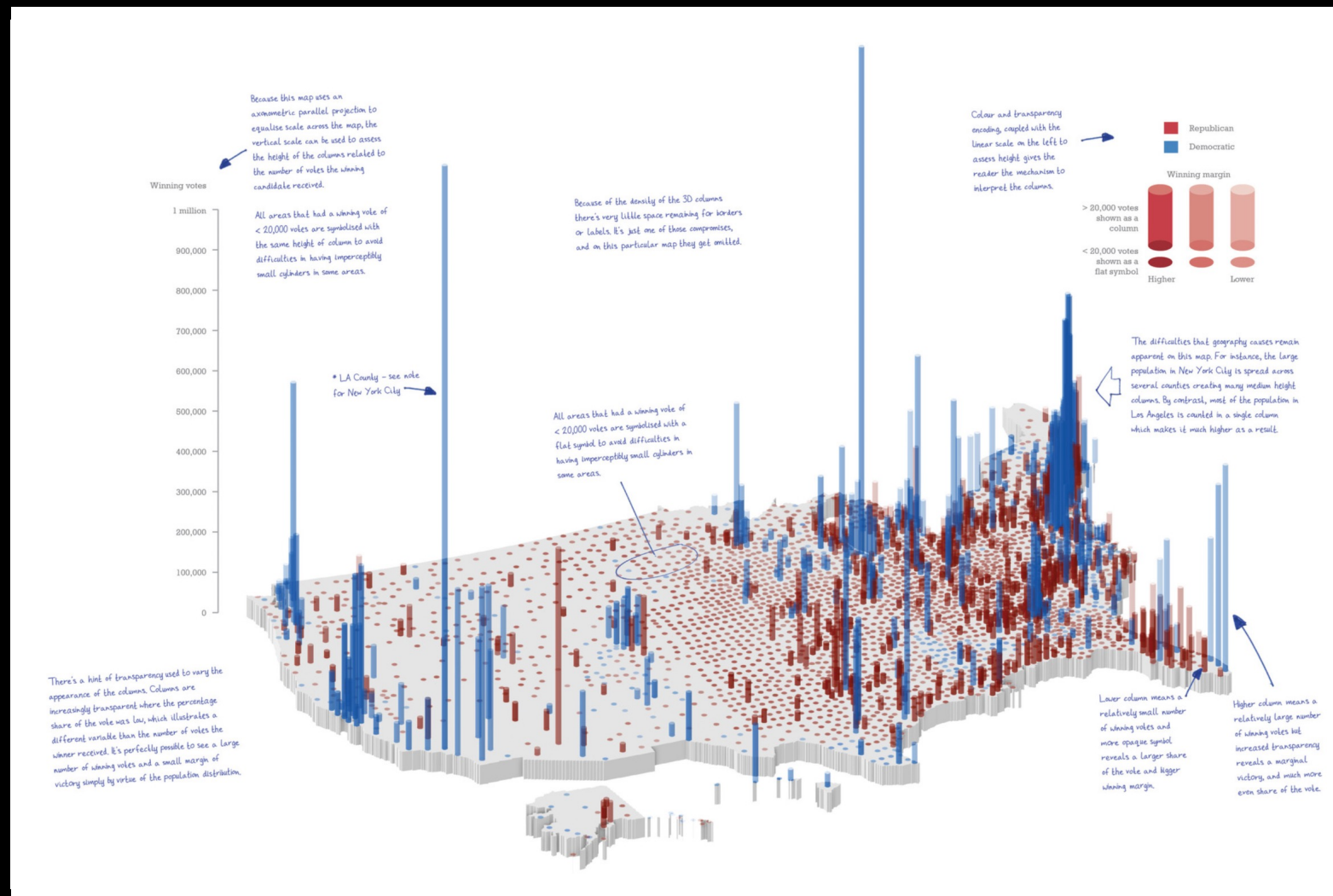
Field (2022)

Triangulated irregular networks (TIN)



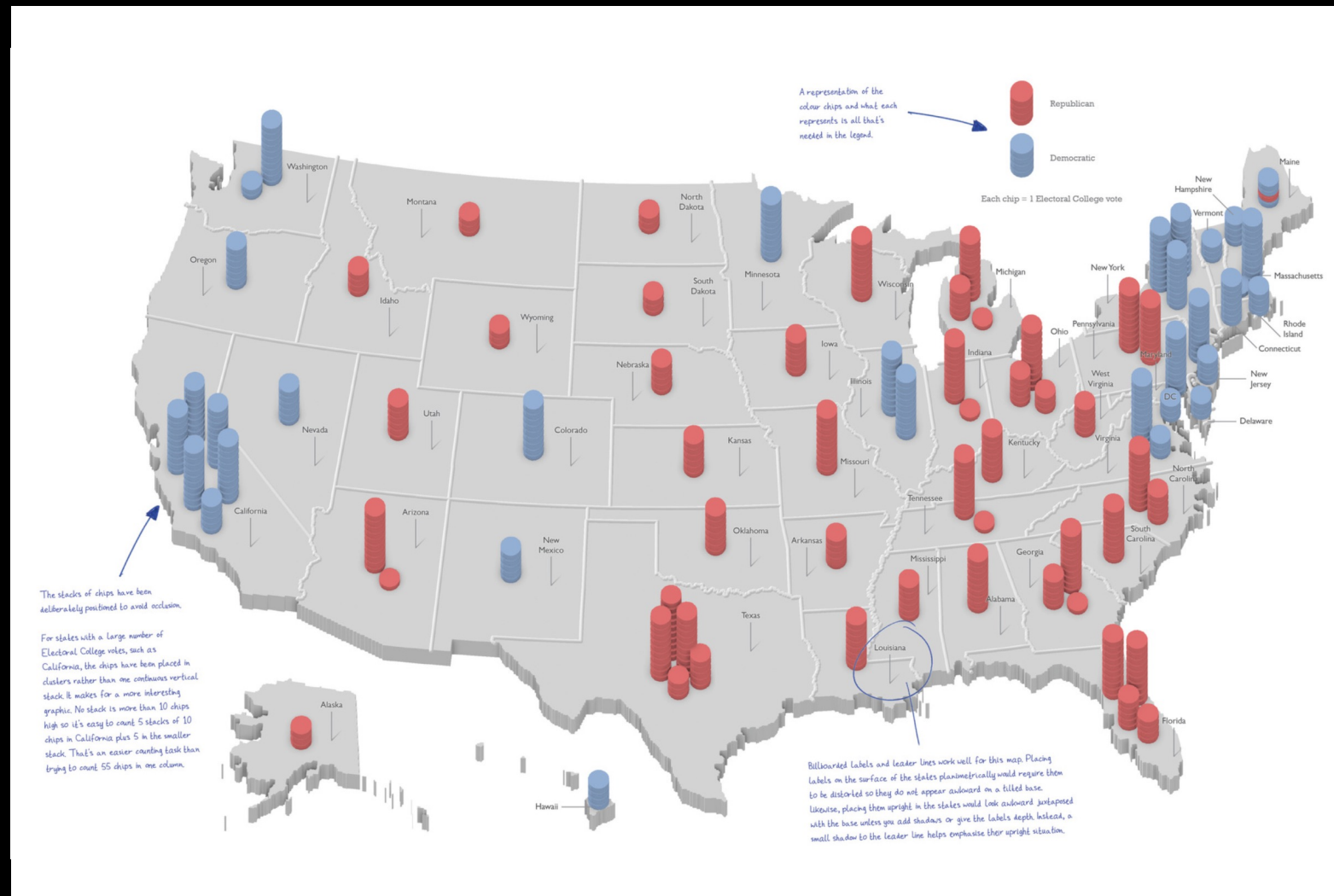
Field (2022)

Illuminated 3D columns



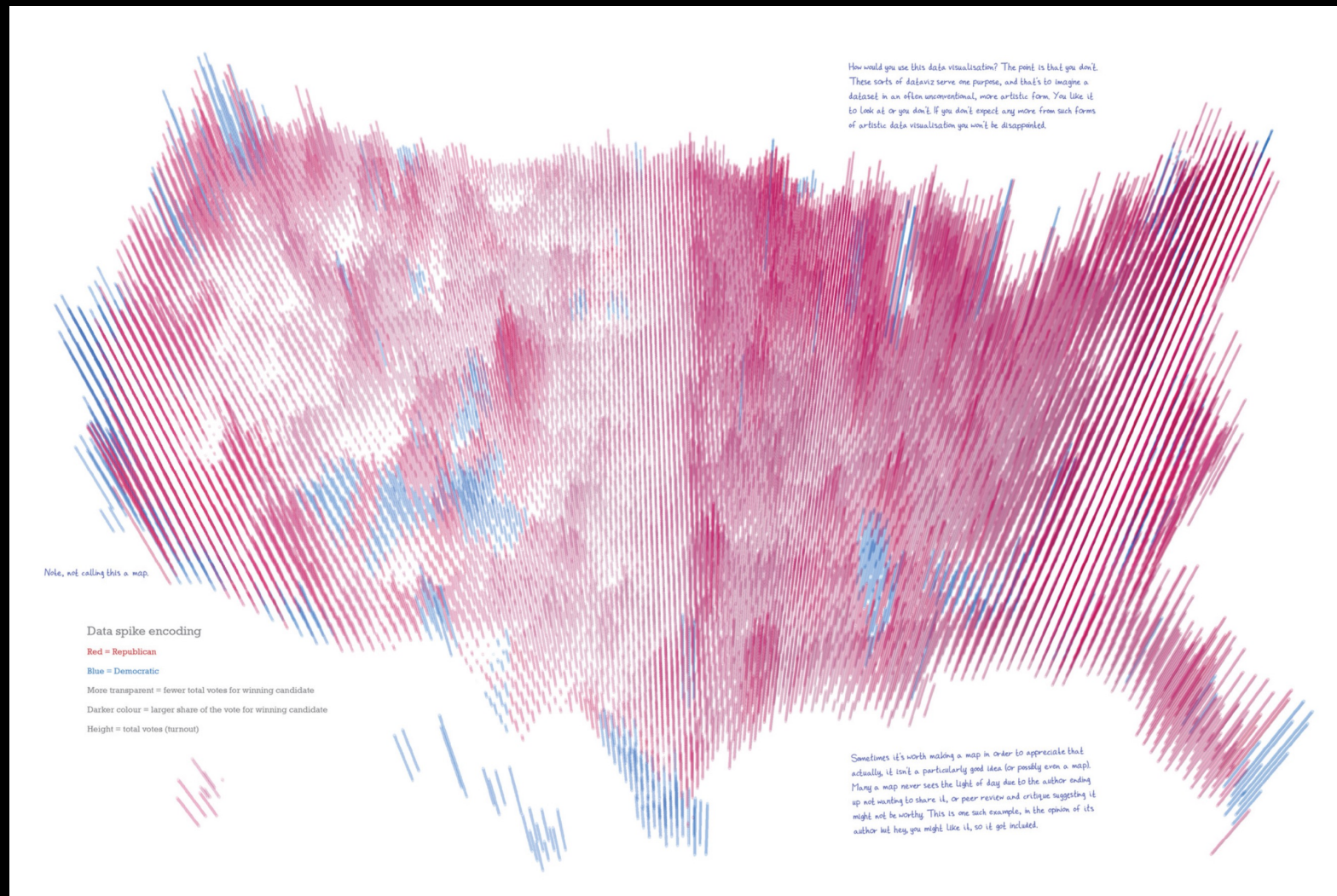
Field (2022)

Stacked chips



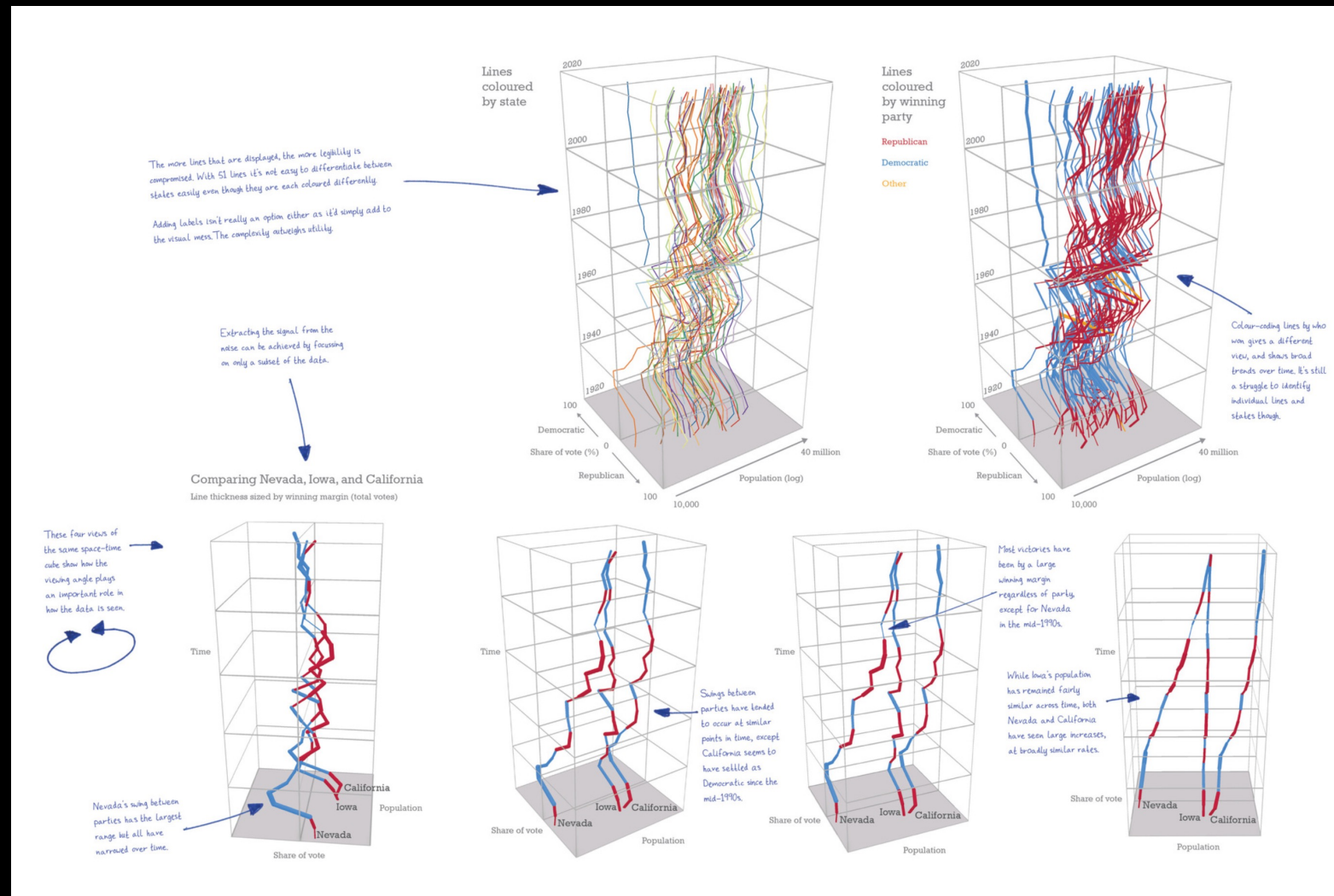
Field (2022)

Data spikes



Field (2022)

Space-time cubes



Field (2022)

